OpenERP User Guide

Enterprise Resource Planning & CRM System

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Chapter 1

OpenERP User's Guide

> **Audience:** End users, system administrators, and business analysts operating an OpenERP installation. For installation and server setup, see INSTALL.md.

1.1 Table of Contents

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1.2 1. Getting Started

1.2.1 1.1 First Login

Navigate to `http://<server>:5173` in a browser. Log in with the credentials provided by your administrator. The default admin account created at install time is `admin` / `admin` — **change this password immediately** under Admin → Users before going live.

The session token expires after 8 hours. You will be redirected to the login screen automatically when it expires.

Collapsing the menu: the □ button at the top of the sidebar collapses it to a narrow icon rail, freeing screen width for the grids. While collapsed, hovering over the rail temporarily expands the full menu over the page; click □ again to pin it open. The choice is remembered per browser.

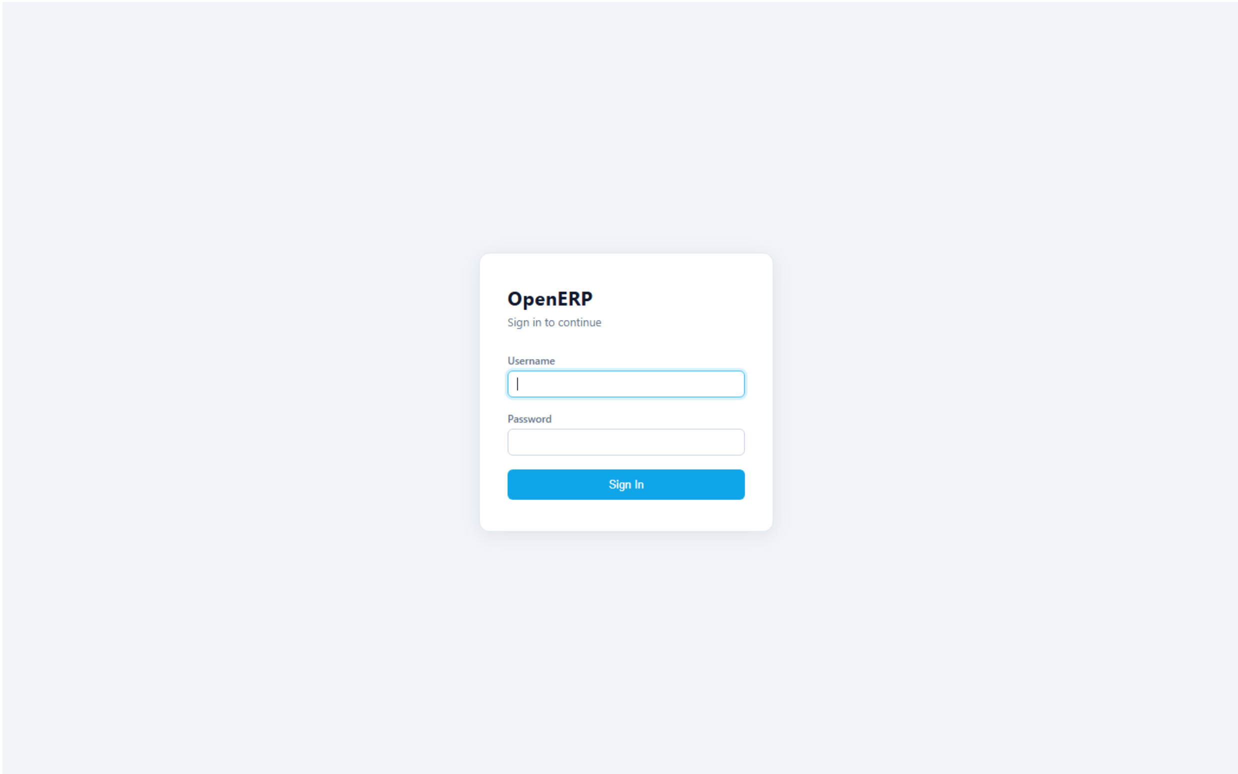


Figure 1.1: OpenERP login screen

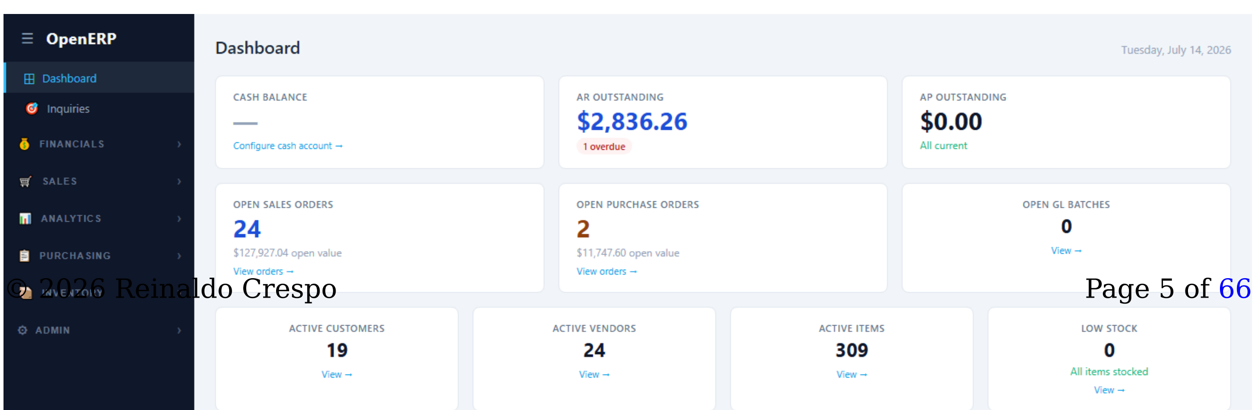
1.2.2 1.2 Dashboard

The Dashboard shows live KPIs pulled from the database:

Tile	What it shows
Cash	Balance of the GL account configured as `gl_cash_account`
AR Balance	Sum of open AR invoice balances
AP Balance	Sum of open AP bill balances
Open SO	Count and value of sales orders in OPEN or PARTIAL status
Open PO	Count and value of purchase orders in OPEN or PARTIAL status
Overdue AR	Invoices past due date

Check the Dashboard every morning. A rising AR Balance combined with a flat Cash balance is an early warning that collections need attention. A rising AP Balance approaching its due dates means a check run is due.

> **Stuck on any screen?** If your administrator has enabled it, a floating □ **AI Help** button in the bottom-right corner answers "how do I…" questions from this guide — see §8.14.



- **Select a row** by clicking it (it highlights light blue). All row actions — Edit, Detail, Ship, Post, Void, and so on — live in the **action bar** directly under the grid and act on the selected row, like a spreadsheet status bar.
- **Column filters (Excel-style)** — columns that show a small ▼ button in the header can be filtered like Excel's AutoFilter: click the button, then check or uncheck values in the list. The popup has a search box for long value lists and a **(Select all)** toggle. Filters on several columns combine — e.g. Status = OPEN *and* Vendor = a specific supplier. A column with an active filter shows its title and ▼ in blue. To clear a filter, re-check **(Select all)**.
- **Date filters** — date columns (Order Date, Invoice Date, Due, Ship By…) have the same ▼ button, but the popup offers **From** and **To** date pickers instead of a checkbox list. Leave one side blank for an open-ended range ("everything since June 1"). **Clear** resets it. Date and column filters combine, so "OPEN orders placed in the last two weeks" is two clicks.
- **Export CSV**, where offered in the page header, downloads the current list.

1.2.3.1 The two kinds of list — and where to type when you're looking for something

This matters, because it tells you *which box to use* to find a record:

Setup and master lists — Items, Customers, Vendors, Chart of Accounts, Warehouses, and most Admin setup tables (Payment Terms, Tax Codes, Customer Classes, Users, Groups…). These load **every record at once**, so there are no page buttons. The **quick search** box above the grid filters as you type, and the ▼ column filters apply to the **entire list** automatically. Nothing is hidden on another page.

Document and log lists — Sales Orders, Invoices, Bills, Payments, POs, Quotes, Requisitions, Transfers, Counts, GL Journals, the Audit/Message/Email logs, and Sales Tax jurisdictions. These grow without limit, so they load one page at a time (page buttons at the bottom). They each have a **Search** box above the grid that searches the **whole table on the server**, not just the page you're looking at — type an order number, a customer or vendor name, a check number, and press Enter. Use that to *find* records; use the ▼ column filters to *refine* what's on screen.

You can always tell which kind you're on at a glance: **page buttons at the bottom means it's a document list**, so reach for the Search box rather than scrolling.

> **In short:** on a document list, the **Search box finds across everything**; the **▼ filters narrow what you're looking at**. On a master list, both cover everything already.
>
> **Tip:** Answering an ad-hoc question is usually two clicks. On the Items list, filter the **Vendor** column to one supplier and the **Type** column to STOCK — the grid now shows exactly that vendor's stocked catalog, and Export CSV gives you the file. On Sales Orders, set the **Order Date** filter to last month and the **Status** filter to OPEN to see what's still outstanding from that period.

1.2.4 1.4 Company Setup

Go to **Admin → Company Settings** to configure:

- **Company name, address, phone, website** — printed on all PDFs
- **Logo** — upload a PNG or JPG; embedded in invoices, statements, SOs, and POs
- **GL account mappings** — tells the system which GL account to debit/credit for AR, AP, cash, revenue, COGS, inventory, and tax payable. These are the `gl_*_account` config keys. Without them the system falls back to the first active account with the matching subtype.

- **PDF document templates** — configurable header subtitle, footer note, and per-document terms text (invoice terms, SO terms, PO terms)
- **SMTP** — for emailing invoices, SOs, and POs directly from the system

  Do this first. Until the GL account mappings are configured, the auto-posting engine posts to the first matching account it finds. If your chart of accounts has multiple accounts of the same subtype, the wrong account may receive the entry. Set the mappings before processing any transactions.

1.2.5 1.5 Chart of Accounts

Before processing any transactions, import or manually create your chart of accounts under **GL → Accounts**.

Each account has: - **Account number** — user-defined (e.g. `1000`, `4000-01`) - **Account name** - **Account type** — ASSET, LIABILITY, EQUITY, REVENUE, EXPENSE - **Account subtype** — AR, AP, CASH, SALES, COGS, INVENTORY, etc. The subtype is used by the auto-posting engine to resolve accounts from the config keys above. - **Is header** — header accounts cannot receive journal line amounts; they exist only for subtotaling in financial statements.

Use **GL → Fiscal Years** to create the current fiscal year, then **GL → Periods** to create the twelve monthly periods inside it. The system will not allow posting to a closed period.

1.2.6 1.6 Payment Terms and Tax Codes

- **Admin → Payment Terms** — define NET30, NET60, 2/10-NET30, etc. Used on customer and vendor records and defaulted onto SOs, POs, invoices, and bills.
- **Admin → Tax Codes** — define tax rates (e.g. TX-STATE at 6.25%). Assigned to items and overridable per invoice line. For US sales tax with multi-jurisdiction support, use the Sales Tax engine (Admin → Sales Tax).

1.2.7 1.7 Go-Live Setup Checklist

Before your first live transaction, confirm the following:

- ☐ Company name, address, and logo are entered in Company Settings
- ☐ GL account mappings (all `gl_*_account` keys) are set
- ☐ Fiscal year and all 12 periods are created
- ☐ Chart of accounts is imported (Admin → Import Tool → Chart of Accounts) or built by hand
- ☐ Payment terms are created and assigned to customers and vendors
- ☐ Tax codes or sales tax jurisdictions are configured
- ☐ Item master is loaded — categories, UOM, warehouses/bins, then items, then price lists (all importable, in that order — see §8.12)
- ☐ Customer master is loaded with credit limits and terms (importable)
- ☐ Vendor master is loaded with 1099 flags and terms (importable), then the vendor item catalog if used
- ☐ Admin user password changed from default
- ☐ Non-admin user accounts and groups are created
- ☐ Warehouse and bin locations are created (importable)
- ☐ Opening inventory balances are entered via manual adjustments
- ☐ Open AR invoices are entered or imported (Admin → Import Tool → AR Invoices — does not post to GL)
- ☐ Open AP bills are entered or imported (Admin → Import Tool → AP Bills — does not post to GL)

- ☐ In-flight Sales Orders / Purchase Orders are imported if migrating mid-cycle (Admin → Import Tool)
- ☐ Opening GL balances are imported (Admin → Import Tool → GL Opening Balances) — do this LAST, after AR/AP invoices/bills are loaded, since those import types don't post to GL themselves

---

1.3 2. How the Modules Connect

Understanding how data flows between modules prevents duplicate work and explains why a transaction in one area shows up somewhere unexpected.

1.3.1 2.1 The Core Flows

```
PURCHASING (PO)                    SALES (SO)
     |                                  |
     | Receive goods                    | Ship goods
     v                                  v
INVENTORY (on-hand + FIFO layers) <------+
     |                                  |
     | COGS consumed on shipment        |
     v                                  |
GENERAL LEDGER (auto-posted)       AR INVOICES
     ^                                  |
     |                                  | Customer pays
     |                                  v
AP BILLS <-- (bill from PO)       AR PAYMENTS
     |                                  |
     | Vendor paid                      | Applied to invoice
     v                                  v
AP PAYMENTS -----> GL: DR AP / CR Cash   GL: DR Cash / CR AR
```

Every transaction that touches money posts a GL batch automatically. You do not need to create manual journal entries for normal operations.

1.3.2 2.2 Auto-Posting: What Posts to GL and When

The following events create a GL journal batch silently in the background the moment you save:

Event	GL Debit	GL Credit
AR Invoice saved	AR	Revenue (+ Tax Payable)
AR Payment applied to invoice	Cash	AR
AP Bill saved	Expense	AP
AP Payment applied to bill	AP	Cash
SO Shipment posted	COGS	Inventory
Inventory Adjustment (positive qty)	Inventory	Inv-Adj
Inventory Adjustment (negative qty)	Inv-Adj	Inventory

Event	GL Debit	GL Credit
AR Credit Memo created	Revenue	AR
AP Debit Memo created	AP	Expense
PO Receive (no direct GL)	—	— (FIFO layer created; GL posts when items ship)

Key implication: When you receive a PO, the goods enter inventory and a FIFO cost layer is created, but no GL entry posts at that moment. The GL entry (COGS) posts when those items are later shipped on a Sales Order. This is the standard perpetual inventory model — the inventory asset account increases on receipt via the AP bill (DR Inventory is embedded in the COGS flow), and COGS hits the income statement on shipment.

1.3.3 2.3 The AR-to-Cash Chain

Many users expect cash to appear in their bank account when they “apply” a payment. What actually happens is a GL entry (DR Cash / CR AR). The cash account balance rises, the AR balance falls. The actual bank deposit must be reconciled in **Bank Reconciliation** (§7.8) when the bank statement confirms the funds cleared.

```
Invoice saved       → AR balance rises, Revenue balance rises
Payment saved       → No effect yet (payment is unmatched)
Payment applied     → AR balance falls, Cash (GL) balance rises
Bank statement rec  → Statement line matched to payment, bank confirmed
```

1.3.4 2.4 The PO-to-Expense Chain

```
PO created          → No GL effect
PO received         → On-hand rises; FIFO layer created
AP Bill from PO     → AP balance rises, Expense (or Inventory) balance rises
AP Payment applied  → AP balance falls, Cash balance falls
```

For **inventory items**: the AP bill posts DR Inventory / CR AP — the cost stays in inventory until the items ship, at which point COGS is posted. For **service or expense items** (non-inventory): the AP bill posts DR Expense / CR AP directly.

1.3.5 2.5 How a Sales Order Touches Five Modules

When a Sales Order is created, processed, and invoiced:

1. **SO created** — no financial effect; just a commitment
2. **SO shipped** — inventory on-hand falls; FIFO cost layers consumed; GL: DR COGS / CR Inventory
3. **SO invoiced → AR Invoice** — AR balance rises; GL: DR AR / CR Revenue
4. **AR Payment applied** — AR balance falls; GL: DR Cash / CR AR
5. **Bank Reconciliation** — payment matched to bank deposit; GL cash confirmed

The sales rep’s commission accrues conceptually at step 3 (when the invoice is created). The Commission Report at **Sales → Commissions** shows commission earned on invoiced SOs.

1.3.6 2.6 ATP (Available to Promise) and How Inventory Informs Sales

When you select an item on a Sales Order, the **ATP badge** shows:

```
ATP = On-hand − Committed (open SO lines) + On-order (open PO lines)
```

A green badge means you can promise this quantity to the customer. A red badge means either you will need to back-order or you need to place a PO first. The ATP calculation happens in real time and reflects all open SOs and POs — it is the link between sales commitments and purchasing plans.

1.4 3. Order-to-Cash

The Order-to-Cash workflow moves a sales opportunity from quotation through cash collection.

```
Customer → Quote → Sales Order → Shipment → AR Invoice → Cash Receipt → Applied → Reconciled
```

1.4.1 3.1 Customer Master

AR → Customers — create a customer before entering any orders.

Key fields: - **Customer number** — unique identifier; printed on documents - **Credit limit** — if set, the system blocks SO creation when the customer's open AR + open SO value exceeds this limit - **On credit hold** — manually block all new orders for this customer; a red banner appears on the SO form when a held customer is selected - **Default sales rep** — auto-populated on new SOs when the customer is chosen - **Customer class** — used by matrix pricing (e.g. KEY, STANDARD, PROSPECT) - **Payment terms** — defaulted onto invoices

Ship-to addresses: Click **Ship-To** on the customer row to manage multiple delivery addresses. One can be flagged as default; it auto-selects on the SO form.

Contacts: Click **Contacts** to add named contacts with phone and email. Contacts flagged "primary" receive emailed documents.

Customer Detail (Requirements / Timeline / Documents): Click **Detail** on a selected customer to open the customer's operational record — three tabs that turn the Customer Master into a complete knowledge base rather than a contact list:

- **Requirements** — the customer's standing operational rules, visible to every department: shipping preferences (carrier, freight terms), packaging / labeling / inspection requirements, quality expectations, inventory strategy, and flags for Vendor-Managed Inventory and Stock & Release programs. Enter them once here instead of re-explaining them in emails; anyone shipping or quoting for this customer can look them up.
- **Timeline** — the relationship history: dated, typed entries (Visit, Business Review, Engineering, Quality Issue, Corrective Action, Commercial, Milestone) with a summary and optional details. Add an entry after every customer visit or review — the timeline is permanent (removing an entry hides it but never destroys it) and survives personnel changes.
- **Documents** — file attachments up to 25 MB each: tax-exempt certificates, contracts, NDAs, quality agreements, packaging/labeling specs, routing guides. Certificates carry an **expiration date**; dates within 90 days (or past) show in red so an expired resale certificate never surprises you at invoice time. Removing a document deactivates it; the file stays on disk.

  What to do next after adding a customer: Set the credit limit and payment terms before their first order. A customer with no credit limit will never be blocked by the credit check, regardless of how much they owe. Then open **Detail → Requirements**

and record their packaging/labeling/inspection rules while they're fresh from onboarding.

1.4.2 3.2 Sales Quotes

Sales → Quotes — optional step before creating a firm order.

> For opportunities involving parts that don't exist in the Item Master yet (custom-engineered components needing supplier sourcing first), start from the **Inquiry Master** (§3.10) instead — it can spawn this quote for you once sourcing is done.

Statuses: `DRAFT` → `SENT` → `ACCEPTED` or `EXPIRED` or `CANCELLED`

- **Send** converts a DRAFT quote to SENT (recordkeeping only; no email is sent automatically unless configured).
- **Accept** creates a Sales Order from the quote lines and marks the quote ACCEPTED.
- **Expire** marks the quote EXPIRED; no order is created.

> **What comes next:** After a customer accepts a quote (verbally or in writing), click **Accept** to convert it to a Sales Order. The quote line prices and quantities carry over exactly. You can then edit the SO if quantities need adjustment before shipment.

1.4.3 3.3 Sales Orders

Sales → Orders — the core transaction.

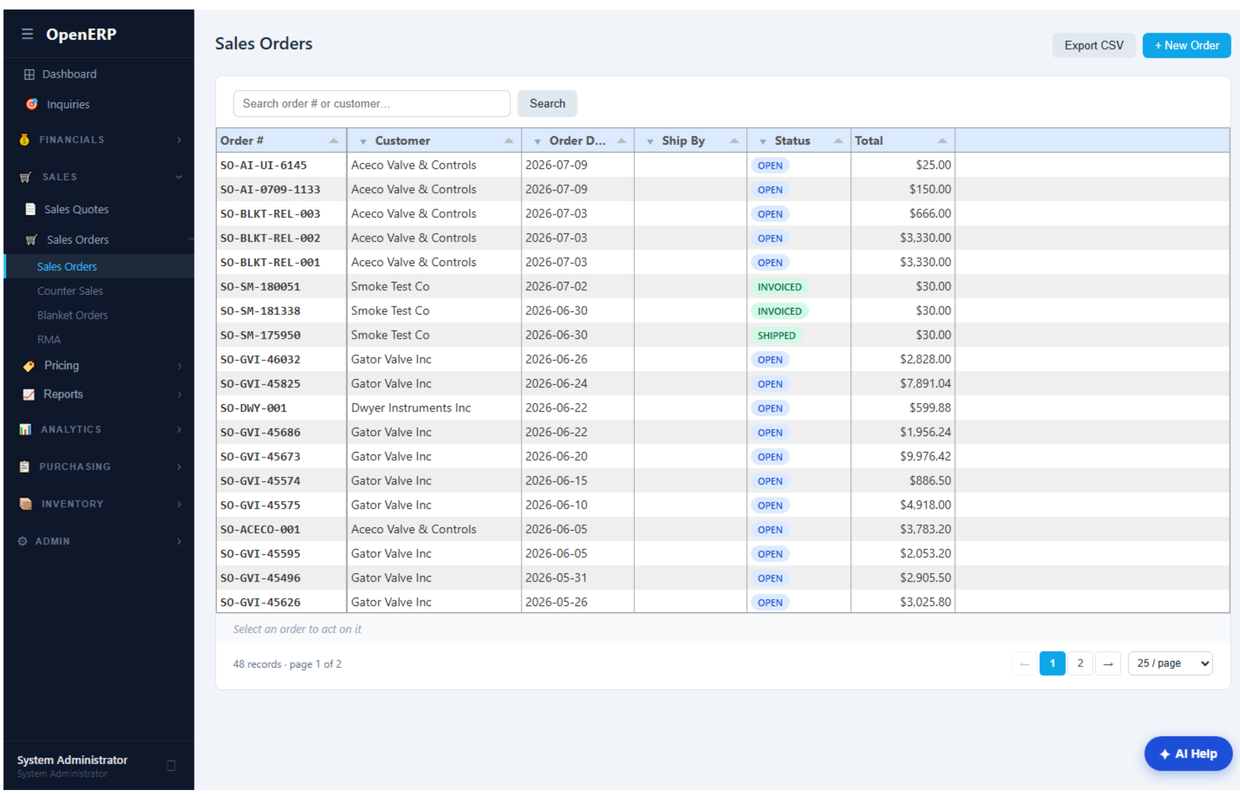


Figure 1.3: Sales Orders list showing order status, totals, and the action bar below the grid

1.4.3.1 Creating a Sales Order

1. Choose a **customer**. The system immediately checks credit: a green/yellow/red banner shows available credit. If the customer is on hold, the **Create Order** button is disabled.
2. Set **order date**, **requested ship date** (optional), **period** (accounting period for GL posting), **payment terms**, **ship-to address**, and **sales rep**.
3. Add **line items**. For each line:
   - Select an **item**. The system fetches the price (checking contract → customer price list → matrix pricing → base price list → list price, in that order) and shows the **ATP** (available-to-promise) badge in green or red.
   - Enter **qty ordered**. If qty breaks apply the price may adjust.
   - Optionally set a **promise date** for this specific line.
   - Check **drop-ship** if the vendor should ship directly to the customer (see §3.7).
4. Save. The order is created in `OPEN` status.

   What comes next after saving a Sales Order: - If ATP is red on any line, go to **Purchasing → Orders** and place a PO for the shortfall. Or go to **Inventory → Reorder Suggestions** to see all understocked items at once. - If the order should ship today, proceed to the **Ship** action (below). - If payment terms are prepaid, go to **AR → Deposits** to record the customer's payment before shipping.

1.4.3.2 Sales Order Statuses

Status	Meaning	What to do
OPEN	No lines shipped yet	Pick and pack; click Ship when ready
PARTIAL	Some lines shipped	Ship remaining lines or create a back-order
SHIPPED	All lines shipped	Click Invoice to generate the AR invoice
INVOICED	AR invoice generated	Monitor payment in AR → Payments
CANCELLED	Voided before fulfilment	No further action needed

1.4.3.3 Shipping

Click **Ship** on an order in OPEN or PARTIAL status. Enter: - **Ship date**, **period**, **warehouse** - Per-line **qty shipped** (defaults to remaining qty) - Optionally: BOL number, carrier pro number, pieces, weight, freight class

The system: - Creates `SHIPMENT` inventory transactions (reduces on-hand) - Consumes FIFO cost layers and calculates COGS - Posts a GL batch: DR COGS / CR Inventory - Updates line statuses; sets header to PARTIAL or SHIPPED

> **After shipping:** Print or email the packing list (PDF button). The order status moves to SHIPPED. If not all lines shipped, status is PARTIAL — you can ship again or create a back-order. When all lines are SHIPPED, the **Invoice** button appears.

1.4.3.4 Invoicing

Once fully shipped (`SHIPPED` status), click **Invoice**. Provide an invoice number, invoice date, due date, and period. The system creates an AR invoice from the SO lines with the exact shipped quantities.

> **After invoicing:** The AR Invoice is created automatically and GL auto-posts (DR AR / CR Revenue). Email the invoice to the customer's primary contact using the

> **Email** button on the invoice. Then monitor payment from **AR → Payments**. If the customer does not pay by the due date, the invoice will appear in **AR → Aging** under the overdue buckets.

1.4.3.5 Back-orders

Click **Back-Order** on a PARTIAL order to split remaining (unshipped) qty into a new linked SO. The original order's lines are reduced to shipped qty; the new order covers the remainder.

> **When to use back-orders vs waiting:** If you expect stock within a few days, leaving the order PARTIAL is fine. If the customer needs their invoice now and stock will take weeks, create a back-order so you can invoice the shipped portion immediately. The back-order report (**Sales → Back-Orders**) shows all open back-orders so nothing falls through the cracks.

1.4.3.6 Drop-Ship Orders

After adding drop-ship lines to an SO, click **Drop-Ship PO**. Choose a vendor, PO number, dates, period, and terms. The system creates a PO with the customer's shipping address; when the vendor ships and the PO is received, the linked SO lines are marked shipped automatically — no warehouse inventory movement occurs.

1.4.4 3.4 AR Invoices

AR → Invoices — invoices can be created standalone or from an SO (§3.3).

Standalone invoice creation: 1. Choose customer, invoice number, dates, period, payment terms. 2. Add lines: item, description, qty, unit price, tax rate, and optionally a jurisdiction override. 3. The system computes subtotal, tax, and total. On save it posts a GL batch: DR AR / CR Revenue (+ CR Tax Payable if tax > 0).

1.4.4.1 Voiding an Invoice

Only `OPEN` invoices can be voided. Click **Void**. The system posts a reversing GL batch.

> **When to void vs write-off:** Void an invoice when it was created in error (wrong customer, wrong amount, duplicate). Write-off an invoice when the goods were delivered, the invoice is legitimate, but the customer will not pay. A void removes the revenue from your books; a write-off moves the balance to Bad Debt Expense.

1.4.4.2 Write-offs

For uncollectable balances, click **Write-Off** on an OPEN or PARTIAL invoice. Enter the write-off amount. The system creates an `ar_write_offs` record and posts GL: DR Bad Debt Expense / CR AR.

1.4.5 3.5 Cash Receipts

AR → Payments — record a cash receipt: 1. Choose customer, payment number, date, amount, payment method (CHECK, WIRE, ACH, etc.), and period. 2. Save. The payment is unmatched (status: OPEN). 3. Click **Apply**. Select open invoices and enter the amount to apply to each. The system updates invoice balances and posts GL: DR Cash / CR AR.

> **What comes next after applying a payment:** - The invoice balance falls (possibly to $0, which marks it PAID). - The Cash GL account rises. - The physical deposit

must still be matched to your bank statement in **Bank Reconciliation** (§7.8) when the statement is available. - If the customer took an early-pay discount (e.g., 2/10-NET30) and paid less, apply their actual amount and enter the discount on the apply screen. The remaining balance is posted as discount taken.

1.4.5.1 Credit Memos

AR → Credit Memos — issue a credit for a return or price adjustment: 1. Create with customer, amount, and description. Posts GL: DR Revenue / CR AR. 2. Apply to an open invoice using **Apply** — reduces the invoice balance. 3. **Void** a credit memo to reverse it (GL reversal posted).

> **Credit memo vs voiding the invoice:** If the customer received goods and is returning some of them, issue a credit memo — the original invoice was correct when issued. If the invoice itself was wrong (wrong price, wrong customer), void the invoice and reissue.

1.4.6 3.6 Customer Deposits

AR → Deposits — record a prepayment before the invoice is raised: 1. Create deposit: customer, amount, period. 2. Apply to an invoice when it is ready: GL moves the deposit from Deposits liability to AR. 3. **Void** reverses the application or the original deposit.

> **Procedure for prepaid customers:** Record the deposit as soon as you receive the check or wire. Ship and invoice as normal. When applying the deposit to the invoice, the invoice balance will drop to zero (or to the remaining amount if the deposit was partial). This keeps your AR aging accurate — the deposit does not show as an open invoice.

1.4.7 3.7 AR Aging and Collections

AR → Aging — shows open balances bucketed by 0–30, 31–60, 61–90, 91+ days past due.

> **How to run collections from the Aging screen:** 1. Run the Aging report weekly (or daily for high-volume operations). 2. Focus the 61–90 and 91+ columns first — these are the most at risk. 3. For each overdue customer, click through to the customer record to see the contact name and phone. 4. Log your contact attempt in **AR → Collections** with notes. 5. Customers who are 90+ days with no response: discuss write-off with management. 6. Customers who promise to pay: set a follow-up date in the Collections log.

AR → Collections — a prioritized queue of customers with overdue balances. Log contact attempts with notes; the contact history is visible per customer.

1.4.8 3.8 Customer Statements

AR → Statements — view a customer's transaction history for any date range. **Print** generates a PDF. **Email** sends it to the customer's primary contact.

> **When to send statements:** Send monthly statements to all customers with open balances on the last business day of the month. Many customers use the statement (not individual invoices) to schedule their payments. A statement shows all open invoices, any credits, and the total balance due.

1.4.9 3.9 Counter Sales

Sales → Counter Sales — one-screen entry for walk-in customers: creates the sales order, ships it, invoices it, and records the payment in a single step.

1. Pick the customer, enter SO #, date, and period; add line items — prices auto-fill from the customer's price list, and totals update live.
2. In the **Payment** panel, choose the method (CASH / CREDIT_CARD / CHECK), enter invoice # and payment #, and the amount tendered — **Change Due** is computed automatically.
3. **Complete Sale** runs the whole chain (SO → ship → invoice → payment → apply) and shows the resulting invoice and payment numbers. **New Sale** clears the screen for the next customer.

The Order Details and Payment panels sit side by side on a wide screen; on a narrower window the Payment panel moves below Order Details — the page never scrolls sideways, so the Complete Sale button is always reachable.

1.4.10 3.10 Inquiry Master

Inquiries (top-level sidebar link) — tracks a customer opportunity from the moment it arrives, *before* a part number or quote exists. Traditional flows in this guide start from an existing item; the Inquiry Master starts from a request — often for a custom-engineered part you have never sourced — and carries it through supplier sourcing and customer quoting to a win or a loss. Either way, the record is preserved: **lost inquiries are never deleted**, so the sourcing knowledge (who quoted, at what price, why the business was lost) stays searchable for the next opportunity.

Statuses: `DRAFT` → `QUOTING` → `AWARDED` or `LOST` or `CANCELLED`

1.4.10.1 Creating an Inquiry

1. Click **+ New Inquiry**. Enter the inquiry #, **customer** (contact and sales rep optional), **date received**, and target quote date. Industry, application, and estimated annual volume are free-form context for later reporting.
2. Add **lines**. Each line describes one part being requested:
   - Leave **Existing Item** at "— new part —" when the part doesn't exist yet, and fill in the engineering identity instead: **customer part #**, description, **drawing #**, **revision**, and **material spec**.
   - Or select an existing item for repeat business — description and UOM auto-fill.
   - **Target price** records what the customer hopes to pay; it pre-fills the quote and award forms later.
3. Save. The inquiry is created in `DRAFT`.

The **Notes (Operational Intelligence)** box on the Overview tab is for contextual knowledge that doesn't fit a field — "customer prefers castings from Taiwan", "drawing rev B expected next month". It is editable any time before award/loss.

1.4.10.2 The Engineering Package (Documents tab)

Upload drawings, spec sheets, CAD models, material certs, and customer correspondence on the **Documents** tab (max 25 MB per file). These files are the technical foundation suppliers quote against. Download or remove them from the same tab; removed documents are deactivated, not destroyed.

1.4.10.3 Sourcing: Spawn RFQ and Spawn Quote

- **Spawn RFQ** creates a supplier RFQ (§4.3) pre-loaded with the inquiry's lines. Work the RFQ as usual — send to vendors, record quotes, compare prices in the comparison grid. The inquiry's **Sourcing** tab shows the RFQ's status and vendor-quote count with a **View RFQ →** link straight into it.
- **Spawn Quote** creates a customer Sales Quote (§3.2) from the inquiry's lines. Enter your selling price per line (pre-filled from target price); the quote then follows its normal life — send, accept into an SO, or expire. The Sourcing tab links to it the same way.

Either action moves the inquiry to `QUOTING`. An inquiry can have at most one RFQ and one quote.

1.4.10.4 Award or Loss

- **Award** — the customer gave you the business. For each line, either select the existing item or let the system **create a new Item Master record** from the part details. Enter the winning **vendor** and **unit cost**. On award the system:
  - creates the `items` row (type STOCK, standard cost = the awarded cost) for new parts,
  - records the **customer part number** as a CUSTOMER cross-reference on the item (§6.9),
  - adds the winning vendor to the item's **vendor catalog** (§4.11) as preferred, at the awarded cost,
  - links each inquiry line to its item and marks the inquiry `AWARDED`.
- **Mark Lost** — enter the reason (competitor, price, timing…). The inquiry, its lines, engineering package, RFQ, and quote all remain on file. Six months later, when the same drawing comes back around, search the Inquiries list and you have the full sourcing history.

  What comes next after an award: the new item exists in the Item Master, priced and vendor-linked, so the normal flows take over — accept the spawned quote into a **Sales Order** (§3.3) and convert the RFQ's winning quote into a **PO** (§4.3), or create both directly.

1.4.11 3.11 Order Status Tracking & Customer Notifications

For imported/manufactured goods, weeks pass between "order placed" and "order shipped" — and customers call to ask where their order is. This module answers that proactively: each sales order carries a **status timeline** (vendor manufacturing stages, ship date, estimated arrival, exceptions like a customs hold, warehouse arrival), and every new entry can automatically message the customer as a **text message** (SMS or WhatsApp, via Twilio) and/or **email**.

Recording a status. On **Sales → Sales Orders**, select an order and click **Status**. The timeline modal shows the order's history and an entry form:

- **Status** — pick from the fixed list: Order confirmed, In production, Production complete, Shipped, In transit, On hold at customs, Cleared customs, Arrived at warehouse, Ready for delivery, Delivered, Delayed, or a free-form Update/note.
- **Status Date** (defaults to today) and an optional **Est. Arrival** date.
- **Notes** — free text, included in the customer message (e.g. *"vessel departed Ningbo"*).
- **Notify customer** — checked by default. Uncheck it to record an internal-only entry.

Entries are never destroyed — the □ button only hides one from the timeline (it stays in history and the audit log). Statuses and dates can't be edited after the fact; post a new entry instead — the timeline *is* the record.

Who gets notified, and how. Notification is per-customer opt-in, set on the customer form (AR → Customers → Edit): a **Mobile Number** (international format, e.g. `+19547440286`) with a **Notify via Text / WhatsApp** checkbox, and a **Notify via Email** checkbox (uses the customer's email on file).

Messages are queued, not sent instantly. Adding a status entry places the messages in the **outbound message queue** — the entry shows □ *queued* immediately, and the queue worker sends within about a minute, retrying automatically on transient failures (see **Max Send Attempts** in the Twilio settings; default 3, backing off 1 / 5 / 15 minutes between tries). After a text is sent, the system also confirms **actual delivery** with Twilio — so the status you see progresses □ queued → □ sent → □ delivered (or □ failed with the exact provider error). The full history of every message — including its exact text — lives in **Admin → Tools → Message Log** (§8.15), and each order's Status modal shows the same per-entry.

Setup (administrator). Email needs only the existing SMTP settings (§8.10). Text messages are sent through **Twilio** — configure it under **Admin → Setup → Company → Text / WhatsApp Notifications (Twilio)**:

1. **Channel** — **SMS** (the default) sends ordinary text messages and **works immediately** with any SMS-capable Twilio number; no Meta/Facebook account or approval is involved. **WhatsApp** requires your number to be registered as a WhatsApp sender first — a Meta business-verification process done from the Twilio Console — so the practical path is: run on SMS today, switch the channel to WhatsApp later if/when the sender is approved. Everything else (opt-ins, timeline, logging) stays the same when you switch.
2. **Account SID** and **Auth Token** — from the **Account Info** panel on your Twilio Console home page. Use the **Live** credentials — Twilio also offers "Test credentials", which authenticate but cannot send real messages (sends fail with error 20008 or 21606). The token is stored server-side and always displayed masked.
3. **From Number** — your Twilio number in international format (used for both channels). *WhatsApp-sandbox testing:* use `+14155238886` after joining the sandbox from your phone.
4. **WhatsApp Template SID** (optional; WhatsApp channel only) — WhatsApp only allows a business to message a customer *outside a 24-hour reply window* using a **pre-approved template**. Create one in Twilio (Content Template Builder), get it approved, and paste its `HX…` SID here. The template receives four variables: `{{1}}` order number, `{{2}}` status, `{{3}}` date, `{{4}}` ETA/notes. Ignored on the SMS channel — SMS always sends the plain message text (error 63016 on WhatsApp means a template is required).
5. **Save**, then use **Send Test Message** with your own number to confirm the credentials work. Note for Twilio **trial** accounts: SMS can only be delivered to phone numbers you've verified in the Twilio Console (Phone Numbers → Verified Caller IDs).

   US SMS requires A2P 10DLC registration. US carriers require every business sending SMS from a regular local number to register an **A2P 10DLC brand and campaign** (done once, in the Twilio Console under Messaging → Regulatory Compliance). Until Twilio approves the campaign, messages to US numbers are blocked or filtered (commonly Twilio error 30034). This is a carrier rule, not an OpenERP or account problem — configure everything, then start sending once the campaign shows *approved*.

1.4.12 3.12 Customer Budgets

Sales → Customer Budgets — record the sales you forecast for each existing customer per year, then track how the year is actually going.

Edit Budgets view: pick the year, then enter a monthly budget amount per customer in the grid (row totals and column totals update live). Only changed cells are saved; clearing a cell to blank or zero removes it. Saving requires the `sales.pricing.manage` permission.

Performance view (the dashboard): - **Tiles** — total budget for the year (with the YTD portion), actual invoiced revenue with **% of budget reached**, a **run-rate forecast** (actual annualized over the months elapsed — for past years it equals the actual), and overall **margin %**. - **Budget vs Actual by Month** chart — budgeted vs invoiced revenue for all customers combined. - **Per Customer** table — budget, YTD budget, actual revenue, % of budget (green ≥ 100%, amber ≥ 75%, red below), % of the YTD budget, forecast, COGS, margin, and margin %.

How the actuals are computed: revenue is AR invoice totals for the year (voided invoices excluded); COGS is the FIFO cost of SO shipments (which already includes posted landed costs — see §5.6), so **margin % reflects true landed cost**. Customers with revenue but no budget still appear in the table.

The **Customer Dashboard** (Analytics section) shows the current year's Budget vs Actual chart with the % reached / forecast / margin summary and a link to the detail page.

1.5 4. Procure-to-Pay

`Vendor → Requisition → RFQ → PO → Receipt → AP Bill → Payment → Applied → Reconciled`

1.5.1 4.1 Vendor Master

Purchasing → Vendor Management → Vendor Master — the central supplier database. Every PO, bill, RFQ, rebate, and payment references a Vendor Master record: enter each company once and every module reuses it.

Vendor types. Each vendor is typed to reflect its real role in the supply chain:

Type	Meaning
AGENT	Managing office / trading partner you deal with commercially
MANUFACTURER	An actual factory
TRADING_COMPANY	Trading company (may also have factories under it)
LOGISTICS / INSPECTION / WAREHOUSE_3PL / CUSTOMS_BROKER	Service providers
UNKNOWN_MFG	Placeholder for a factory whose identity is not yet known

Agents and manufacturers. An **agent** is the commercial front door: specs and RFQs go to the agent, the agent decides which of its factories builds the part, and POs/bills/payments are placed with the agent. Each manufacturer record carries an **Agent** field pointing at its agent; a manufacturer with no agent is a **direct** relationship. The Vendor Master grid shows an **Agent column with an Excel-style filter (§1.3)**, plus an agent dropdown above the grid, so you can pull up any agent's stable of factories in one click.

Agents page. Purchasing → Vendor Management → Agents lists every agent with its manufacturer count. Select an agent to see its factories, **Link** an existing vendor to the agent, or **Unlink** one. Create new agents here (or retype an existing vendor as AGENT in the Vendor Master edit form).

Status. Every vendor carries a status: `APPROVED` → normal; `CONDITIONAL` / `PROBATION` → visible warnings; `INACTIVE` → not used for new business; **`BLOCKED` → the system refuses to create**

new purchase orders for this vendor (existing POs and bills are unaffected, so you can still receive and pay what is already committed).

Vendor detail tabs. Select a vendor and click **Detail**: - **Overview** — commercial profile: trade name, website, incoterms, year established, employees, lead-time notes, plus **Internal Notes** visible only to your team (highlighted yellow) - **Contacts** — multiple contacts per vendor by department (Sales, Engineering, Quality, Production, Logistics, Accounting, Management) with email, phone, **WeChat and WhatsApp**; one contact is starred as primary - **Capabilities** — check off what the vendor can do from a controlled list of 28 manufacturing processes grouped by family (Casting, Forging, Machining, Forming, Molding, Finishing, Testing), plus the **materials** they work in (stainless, duplex, super duplex, titanium, …). Because these are structured checkboxes rather than free text, they power the sourcing filters described below - **Certifications** — ISO/API/PED/ASME and other certificates with number, issuing body, and issue/expiry dates. Expiry dates within 90 days show in red - **Documents** — upload audit reports, factory photos, NDAs, banking information, quality manuals and more (25 MB per file, typed from a controlled list, with optional expiry and notes). Download returns the original file; Remove keeps the file archived on the server but hides it from the list - **Performance** — six live metrics computed from actual transactions (POs, total spend, fill rate, on-time delivery, average lead days, defect rate — the same math as the Vendor Scorecard, §4.13) alongside **manual scores** (Quality / Delivery / Commercial, 0–100) that you enter based on judgment; the Overall score is their average, color-coded green/amber/red. Use the manual scores to record what the numbers can't see — responsiveness, engineering support, how they handle problems - **Factory** (manufacturers only) — factory code, plant/quality/production managers, time zone, language, and free-form capacity/equipment/quality notes - **Manufacturers** (agents only) — the factories linked to this agent

Finding the right supplier. Above the grid, the **Any capability** and **Any material** dropdowns filter the list server-side — combined with the Type/Status/Agent/Country column filters, questions like *"approved stainless investment casters under agent X"* are a few clicks. The grid's **Capabilities** and **Certs** columns show profile completeness at a glance, so you can spot vendors whose knowledge base still needs filling in.

Certificate expiry warnings. An amber banner at the top of the Vendor Master lists every certification that is expired or expiring within 90 days, so renewals can be requested from the vendor before an audit catches the gap.

Unknown manufacturers. When you buy through an agent but don't yet know which factory produces the part, create a vendor typed `UNKNOWN_MFG` under that agent (e.g. "Unknown Manufacturer (Agent X)"). When the factory is later identified, simply rename and retype that same record to MANUFACTURER — its ID never changes, so all purchasing history stays attached.

Financial fields (also editable under **AP → Vendors**, the accounting view of the same records): - **Tax ID + Is 1099** — populated on the 1099 report at year end - **Currency** — if not USD, bills are entered in the vendor's currency; FX gain/loss is computed when payment is applied - **Payment terms** — defaulted onto bills

Vendor bank accounts — under **AP → Vendor Bank Accounts**; used for ACH/EFT export.

> **What to do next after adding a vendor:** Set the vendor type and, for manufacturers, the agent. Set payment terms and, if applicable, the currency. If the vendor is a US sole proprietor or LLC who provides services (not goods), check the **Is 1099** box and enter their Tax ID (EIN or SSN) — you will need this at year end. Add at least one contact per department you actually talk to — future RFQs, quality issues, and shipment questions all start from this list.
>
> **Loading many vendors at once:** don't key them by hand — build the vendor spreadsheet described in **§8.12.6** (one row per company, agents and factories in the same sheet) and load it through Admin → CSV Import.

1.5.2 4.2 Purchase Requisitions

Purchasing → Requisitions — internal request before a PO is issued.

Statuses: `DRAFT` → `PENDING` → `APPROVED` / `REJECTED` → `CONVERTED` or `CANCELLED`

- **Submit** sends a DRAFT to PENDING for approval.
- **Approve** (requires system.users.manage or an approver role) advances to APPROVED and stamps `approved_by`.
- **Convert** turns an APPROVED requisition into a PO; choose vendor, PO number, dates, period, and terms in the modal.

> **When to use requisitions:** In organizations where purchasing approval is required before committing spend, requisitions are the gate. The requesting employee creates a DRAFT, submits it, and a manager approves or rejects. Only after approval can a PO be created. For small purchases below a threshold (e.g., under $500), you may configure your policy to allow POs directly without a requisition.

1.5.3 4.3 RFQ and Quote Comparison

Purchasing → RFQs — request quotes from multiple vendors for the same items.

1. Create an RFQ with line items (item, description, qty).
2. **Send** changes status to SENT.
3. Use **Record Quote** to enter each vendor's response (price per line, lead time, availability). When the quoting vendor is an **agent**, a **Proposed Factory** dropdown appears listing that agent's manufacturers — record which factory the agent proposes to build at.
4. The comparison grid highlights the lowest price per line in green; a □ line under the vendor's name shows each quote's proposed factory, so you can compare not just prices but *who would actually make the part*.
5. **Select** the winning vendor's quote.
6. **Convert to PO** creates the purchase order at the quoted prices. The winning quote's proposed factory is carried onto the PO as its factory attribution (§4.4).

> **When to use RFQs:** Use RFQs for any purchase where you have multiple potential vendors, the amount is significant, or you do not have an existing contract price. The comparison grid makes the decision defensible and documented. For routine reorders of items with a Vendor Catalog price (§4.10), you can skip the RFQ and go directly to a PO.

1.5.4 4.4 Purchase Orders

Purchasing → Orders — the core procurement transaction.

Creating a PO is similar to SO creation: choose vendor, dates, period, terms, warehouse, and add lines (item, qty, unit cost). The PO starts in `OPEN` status.

1.5.4.1 Factory Attribution (agent POs)

When the selected vendor is an **agent** (Vendor Master §4.1), a **Factory (built by)** dropdown appears on the form, listing the manufacturers linked to that agent. The PO's commercial party stays the agent — the agent gets the order, the bill, and the payment — but the factory field records *which plant actually builds the parts*. It appears as a **Factory** column on the PO list and can be set or corrected later via **Edit** (e.g., when the agent tells you after the fact which factory got the job). POs converted from an RFQ inherit the winning quote's proposed factory automatically. Leave it blank if the factory isn't known yet.

1.5.4.2 PO Statuses

Status	Meaning	What to do
OPEN	No lines received yet	Wait for delivery; expedite if overdue
PARTIAL	Some lines received	Receive remaining lines when they arrive
RECEIVED	All lines received	Click Bill to create the AP bill
BILLED	AP bill generated	Pay the bill in AP → Payments
CANCELLED	Voided before receipt	No further action needed

1.5.4.3 Receiving Goods

Click **Receive** on an OPEN or PARTIAL PO. Enter receipt number, receipt date, period, and warehouse (+ optional bin). The system: - Creates `RECEIPT` inventory transactions (increases on-hand) - Pushes a FIFO cost layer for each item received - Updates line statuses; sets header to PARTIAL or RECEIVED

> **Receiving at the dock with a barcode scanner:** the desktop Receive button receives all remaining lines in full. For scan-driven receiving — partial quantities, per-line bin placement, and lot capture with labels printed at the dock — use **Warehouse Mode → □ Receive** on a phone or scan terminal (§5.13, "Receiving with the scanner"). Both paths post the same transaction (inventory, FIFO, GL, PO status).
>
> **After receiving:** The goods are now in inventory. The on-hand count on Items list will show the increase. The vendor's bill has not yet been recorded — that happens when you click **Bill**. Do not click Bill until you have the vendor's invoice in hand and have matched it against what you actually received (three-way match).

For drop-ship POs (ship_to = customer address), receiving does not touch inventory — it marks the linked SO lines SHIPPED instead.

1.5.4.4 Creating an AP Bill from a PO

Click **Bill** on a RECEIVED or PARTIAL PO. Enter bill number, bill date, due date, period, and terms. The system creates an AP bill from the received quantities. Three-way matching is tracked (PO qty vs received qty vs billed qty) and visible under **Purchasing → 3-Way Match**.

> **Three-way match procedure:** Before clicking Bill, verify: 1. **PO quantity** matches what you ordered (PO line qty) 2. **Received quantity** matches what actually arrived (receipt qty) 3. **Vendor invoice quantity and price** match the PO
>
> The **3-Way Match** report (Purchasing → 3-Way Match) flags discrepancies. Do not pay a bill with an over-billed quantity or a price higher than the PO without a manager's approval and a PO amendment.

1.5.5 4.5 AP Bills

AP → Bills — bills can be created standalone or from a PO (§4.4).

On save, the system posts GL: DR Expense / CR AP. For foreign-currency bills the USD equivalent is computed using the exchange rate at bill date.

Void — OPEN bills only. GL reversal posted.

> **Standalone bills** (not from a PO) are appropriate for utilities, rent, subscriptions, and professional services — expenses that do not go through the PO/receiving process. For goods that were purchased and received, always create the bill from the PO to preserve three-way match tracking.

1.5.6 4.6 AP Payments

AP → Payments — record a payment to a vendor: 1. Enter payment number, vendor, date, amount, method (CHECK, ACH, WIRE), and period. 2. Click **Apply** to match it against open bills. GL: DR AP / CR Cash. FX gain/loss entries are posted for foreign-currency bills.

Debit Memos (**AP → Debit Memos**) — record a vendor credit (return or price adjustment). Posts GL: DR AP / CR Expense. Apply to open bills to reduce their balance.

> **How the check run works in practice:** 1. On the scheduled check run day (weekly or bi-weekly), go to **AP → Aging**. 2. Review all bills coming due in the next 7–14 days. 3. For each vendor with due bills, create a payment in **AP → Payments** and apply it to those bills. 4. Print checks or initiate ACH transfers outside the system. 5. After the bank confirms the checks cleared, match them in **Bank Reconciliation**.

1.5.7 4.7 AP Aging and Vendor Statements

AP → Aging — open balances bucketed by days past due.

> **Read the aging weekly.** A past-due AP balance means you are damaging your relationship with that vendor and may be incurring late fees. A bill in the 91+ column that you do not recognize may indicate a receiving dispute — pull the three-way match report for that vendor to investigate.

AP → Vendor Statement — shows bills and payments in a date range. Print as PDF or view on screen.

1.5.8 4.8 1099 Report

AP → 1099 Report — lists vendors with `is_1099 = true` and total payments in the selected year. Filter by minimum amount (default $600). Shows a □ badge for vendors missing a Tax ID.

> **Year-end 1099 preparation:** In January, run this report for the prior calendar year. For each vendor shown with a □ badge, contact them to obtain their W-9 (Tax ID). 1099-NEC forms must be issued by January 31. See §10.5 (Annual Procedures) for the full year-end checklist.

1.5.9 4.9 Check Register and Positive Pay

AP → Check Register — log issued checks with check number, payee, amount, bank account. Mark as cleared or void.

AP → Positive Pay — download a CSV file of uncleared checks in the required bank format for fraud prevention.

1.5.10 4.10 Vendor Rebates

Purchasing → Vendor Rebates — track rebate agreements (percent or fixed-per-unit) with effective and expiry dates and a minimum purchase threshold.

1. **Compute** — scans PO receipts in a date range and calculates accrued rebate amounts. Skips receipts already accrued.
2. **Settle** — creates an AP debit memo for the total pending rebate and marks accruals settled.

> **When to compute rebates:** Run the Compute function at month end (or whatever period the rebate agreement specifies). The resulting debit memo appears in **AP → Debit Memos** and can be applied against open bills to that vendor. Rebate agreements often have minimum purchase thresholds — the system enforces these automatically.

1.5.11 4.11 Vendor Item Catalog

Purchasing → Vendor Catalog — stores vendor-specific part numbers, lead times, MOQ, packing quantities, and last cost per item. Mark one vendor as "preferred" per item — the preferred vendor's cost auto-fills on PO lines.

You don't have to come to Purchasing to manage this — see §6.11 for the same data reached from the Items screen instead.

1.5.12 4.12 Blanket POs

Purchasing → Blanket POs — standing contracts with a vendor for a total committed quantity at a fixed price.

Release creates a standard PO for a chosen portion of the committed quantity. Released quantities are tracked against the blanket.

1.6 5. Inventory

Inventory in OpenERP is a perpetual system: every transaction (receipt, shipment, adjustment, transfer) updates on-hand in real time. There is no end-of-day batch update.

1.6.1 5.1 Warehouses and Bins

Inventory → Warehouses — create warehouses (e.g. MAIN, EAST, OVERFLOW). Each warehouse has: - Warehouse code and name - **Bins** — aisle/rack/shelf/bin-code locations within the warehouse; managed via the **Bins** modal on the warehouse row

Warehouse and bin are selectable on PO receive, inventory adjustments, and warehouse transfers.

> **Bin strategy:** Create bins to match your physical layout. Common patterns are A01-01 (aisle A, rack 1, shelf 1). Assigning items to specific bins (via the preferred bin on the item) enables directed putaway and pick instructions. If your operation does not require bin-level tracking, leave the bin field blank — warehouse-level tracking is sufficient.

1.6.2 5.2 On-Hand Quantity

On-hand is calculated in real time as the signed sum of all inventory transactions:

Transaction type	Effect
RECEIPT	+qty (PO receive)
SHIPMENT	−qty (SO ship)
ADJUSTMENT	±qty (manual)
TRANSFER_OUT	−qty (source warehouse)
TRANSFER_IN	+qty (destination warehouse)
RETURN_IN	+qty (RMA receipt)

1.6.3 5.3 Manual Adjustments

Inventory → Items — click **Adjust** on an item row. Enter qty (positive = increase, negative = decrease), date, period, warehouse, optional bin, and notes. The system posts a GL batch: DR Inventory / CR Inv-Adj (or reversed for negative adjustments).

> **When to use manual adjustments:** Use adjustments for: - Correcting a count error discovered during a cycle count - Recording spoilage, damage, or shrinkage - Loading opening inventory balances at go-live - Adjusting for goods that were consumed internally (samples, internal use)
>
> Always enter a meaningful note in the notes field — the Audit Log will show who made the adjustment and when, and the note explains why. Unexplained adjustments are an audit risk.

1.6.4 5.4 Warehouse Transfers

Inventory → Transfers — move stock between warehouses.

1. Create a transfer: from-warehouse, to-warehouse, optional from/to bins, and line items with qty.
2. **Post** executes the transfer: creates TRANSFER_OUT at the source and TRANSFER_IN at the destination.
3. **Cancel** voids an unposted transfer.

> **In-transit inventory:** Once you post a transfer, the source warehouse shows reduced on-hand immediately. There is no "in-transit" status — the destination warehouse shows the increase at the same moment. If your physical transfer takes days (inter-city shipment), you may want to record the transfer only when the goods physically arrive at the destination.

1.6.5 5.5 FIFO Cost Layers

Every PO receipt pushes a cost layer (qty_received, unit_cost, receipt_id, lot_id). When items are shipped via SO, the system consumes the oldest layers first (FIFO). The resulting COGS is posted to GL.

> **Why FIFO matters:** FIFO means the cost on your income statement reflects the oldest inventory you hold. In an inflationary environment (where costs are rising), FIFO produces a lower COGS and higher gross profit than LIFO. This is standard for most distributors and is required under IFRS. The **Valuation (FIFO)** report shows you exactly what your inventory is worth at any point in time.

Inventory → Valuation (FIFO) — shows current FIFO layers by item and warehouse: qty remaining, average cost per unit, and total value. A summary endpoint returns the grand total.

1.6.6 5.6 Landed Costs

Purchasing → Landed Costs — allocate additional costs to PO receipts, the "bump-up" step performed on an import cost worksheet. Component types cover the full worksheet charge list: ocean freight, customs duty, tariff, brokerage, inspection fee, trucking, handling, W/H handling, HMF, ISF filing, marine insurance, insurance, other.

1. Create a landed cost shipment linked to one or more PO receipts.
2. Enter cost components by type and amount. **Each component has its own allocation method** — By Value, By Weight, or By Quantity — and an optional **Payee / Ref #** (the carrier, broker, or document number, e.g. "OEC SS26055797"). Ocean freight is typically allocated By Weight; misc charges and inspection fees By Value, matching how an FTZ bump-up worksheet spreads each charge column.
3. For tariff costs, click **Compute from item tariff rates** to auto-calculate from the items' `tariff_rate` field.
4. Click **Compute Bump-Up** to see the per-item worksheet before posting: qty, weight, PO unit cost, invoice cost, each charge component's per-item share, total added charges, **adjusted unit price**, total cost, and **% of PO cost** — plus a variance line confirming the charges entered equal the amounts spread to items.
5. **Post** distributes each component down to the receipt's item lines by that component's method and increases the FIFO layer unit cost per item to the adjusted unit price shown in the bump-up view. GL entries are not generated at this stage — the adjusted cost flows into COGS when items are shipped.

   By Weight requires item weights. Weight allocation uses `weight_lbs` from the Item Master (Items → item form, Freight/Dims fields). If none of the received items have a weight, saving a By Weight allocation is rejected with a message telling you to set the weights or switch to By Value / By Quantity.

   Why landed cost matters: If you import goods, the unit cost on the PO is typically just the supplier's price. The true landed cost includes freight, customs duties, and broker fees — often adding 10–25% to the PO price. Without landed cost allocation, your FIFO layers understate the true cost of your inventory, and your COGS is understated. Post landed costs before you ship those items; once a FIFO layer is consumed by a shipment, the layer is gone and cannot be retrospectively adjusted.

1.6.7 5.7 Physical Inventory Count

Inventory → Physical Count — a formal full-warehouse count.

1. **Create** a count session for a warehouse. The system snapshots current on-hand as `system_qty` for every active item.
2. Enter `counted_qty` for each line.
3. **Post** creates ADJUSTMENT inventory transactions for variances and posts GL entries for the variance value.

   Physical count best practices: - Schedule a full physical count once per year, typically at fiscal year end or a slow period. - Freeze all receipts and shipments during the count (lock the warehouse). - Count every item at least twice; have a second person recount any item with a significant discrepancy. - Post the count as soon as possible after completing it to minimize the gap between the snapshot and the true count. - Document and review all variances over a threshold (e.g., $500) before posting — unexplained large variances may indicate theft or systemic errors.

1.6.8 5.8 Cycle Counts

Inventory → Cycle Counts — targeted per-item counts on a rolling schedule.

1. **Schedule** a count for an item, warehouse, and optional bin with a scheduled date.
2. **Enter Count** — update counted_qty.
3. **Approve** — creates an ADJUSTMENT transaction and posts the variance to GL.

Filter by ABC class (A/B/C) to prioritize high-value items.

> **Cycle count program:** A well-run cycle count program eliminates the need for an annual shutdown physical count. The idea: - **A items** (high value, top 70% of COGS): count monthly - **B items** (mid value, next 20%): count quarterly - **C items** (low value, bottom 10%): count semi-annually
>
> Each week, schedule the counts due that week, have warehouse staff count and enter results, and approve any within tolerance. Investigate variances above your threshold (e.g., >2% of value). Over time, cycle counts build much more accurate inventory records than an annual physical count.

1.6.9 5.9 Reorder Suggestions

Inventory → Reorder Suggestions — lists items where on-hand is below the reorder point. Shows reorder qty, min/max qty, standard cost, and reorder value. Filter by warehouse.

> **How to use reorder suggestions:** 1. Check the Reorder Suggestions screen Monday morning (or daily for fast-moving items). 2. For each item, verify the reorder point and reorder qty are still correct (demand patterns change). 3. Group items by preferred vendor (visible in the Vendor Catalog). 4. Create a PO per vendor covering all items from that vendor that need replenishment. 5. The reorder qty shown is the suggested quantity; adjust up if you have a promotion coming or down if demand has slowed.

1.6.10 5.10 Auto-Replenishment

Inventory → Replenishment — click **Run** to auto-generate draft POs for all items below reorder point, grouped by preferred vendor. Review and post the draft POs from **Purchasing → Orders**.

1.6.11 5.11 ABC Classification

Inventory → ABC Classify — ranks all items by annual COGS and assigns A (top 70%), B (next 20%), C (bottom 10%). Updates the `abc_class` field on each item. Run annually or after significant volume changes to keep cycle count scheduling meaningful.

1.6.12 5.12 Demand Forecasting

Inventory → Items → Forecast modal — shows 12-month transaction history and computes 3-month, 6-month, and 12-month moving average daily demand. Includes a projected date when current on-hand will reach the reorder point at the forecast demand rate.

1.6.13 5.13 Warehouse Mode — Barcode Scan / Inquiry

Inventory → Warehouses → □ Scan / Inquiry (or browse directly to `/wh`) — a mobile-first, scan-driven screen designed for a phone or an Android barcode terminal on the warehouse floor. It renders full-screen without the desktop menu; the **Desktop ↗** link in the header returns to the normal application.

Scan (or type) any code into the input box and press Enter. The system identifies what the code is, checking in this order:

Code scanned	What you get
Item number	Item card: on-hand, committed, on-order, and **ATP** tiles; on-hand by warehouse and bin; lots on hand (expiry dates within 90 days highlighted)
Factory UPC / MPN (item cross-reference)	Same item card — the screen notes it matched by cross-reference
Bin code	Bin card: warehouse, aisle/rack/shelf, and current contents. Tap any content row to jump to that item's stock card
Lot or serial number	Lot/serial card with a **View item stock** button
SO / PO / transfer number	Document card with status, date, amount, and an **Open on desktop** link

If a code matches more than one record (the same bin code in two warehouses, the same lot number on two items), a pick list appears. If nothing matches, the screen tells you which lookups it tried. The last five scans appear as tappable chips at the bottom.

1.6.13.1 Receiving with the scanner (Receive tab)

The **Receive** tab turns a PO receipt into a scan-driven dock workflow:

1. **Scan the PO number** (from the barcoded paperwork, or type it). The screen shows the vendor and every open line with ordered / received / remaining quantities. Scanning an OPEN or PARTIAL PO from the Inquiry tab also offers a **Receive this PO →** shortcut.
2. **Scan an item** (item number or factory UPC) — or tap its line. An entry panel opens with the quantity defaulted to what's still open on that line. Adjust for a partial receipt, and optionally enter a **lot number and expiry** (the lot record is created automatically).
3. **Scan the destination bin** while the panel is open. The first bin scanned also sets the receipt's warehouse (or pick the warehouse from the dropdown if bins aren't labeled yet); every later bin must belong to the same warehouse.
4. **Add to receipt** stages the line — the PO list shows a "+n staged" badge. Repeat for the other cartons.
5. **Post receipt** writes everything in one transaction: inventory transactions with bin and lot, FIFO cost layers, PO line/header status, GL — the same posting the desktop Receive uses. The receipt number is generated automatically (`RCV-mmdd-nnnn`).
6. If any lots were captured, the confirmation card offers **lot labels** — PDF sheet or straight to the label printer — so goods are labeled before they leave the dock.

Over-receiving is blocked: a quantity above what's open on the line is rejected both on the entry panel and by the server. Partial receipts leave the PO (and line) in PARTIAL status; scan the same PO next container to receive the rest.

Scanner hardware: any barcode gun that types its scan followed by Enter (a "keyboard wedge" — the default mode of Zebra DataWedge and similar Android terminals, and of USB scanners) works with no configuration: keep the cursor in the scan box and pull the trigger. On Android phones and scan terminals, a camera button also appears — point the camera at any barcode to scan without a gun. See §5.14 for recommended scanners, printers, and label media.

> **Barcode labeling practices** — the scan screen is only as useful as your labels. Before rolling out scanning: - **Label every bin.** Walk the warehouse and settle an aisle–rack–shelf scheme first (e.g. `A-01-01`), create the bins under **Inventory → Warehouses → Bins**, then print a barcode label per bin. Use polypropylene thermal-transfer labels — they last years. - **Every item needs a scannable identity.** The item number is the natural barcode. If goods arrive with factory UPCs, record them as item cross-references (**Items → Xrefs**, type UPC) so either code resolves. - **Pre-print carton labels for unlabeled imports.** Print item labels from the PO *before* the container arrives, and label cartons as they are unloaded. - **Label lots at receiving.** A lot label carrying item, lot number, and expiry date makes later lot lookups a single scan. - **Barcode your paperwork.** Put the SO/PO/transfer number as a barcode on printed documents — one scan pulls up the transaction. - **Nothing moves without a scan.** Every putaway, pick, and relocation should be recorded (see Transfers, §5.4); accurate bins depend on it.

1.6.14 5.14 Barcode Hardware and Label Printing

Everything in Warehouse Mode is driven by two pieces of hardware: something that **reads** barcodes and something that **prints** them. Neither requires special drivers or integration — scanners type into the web page, and label printers accept jobs over the network.

1.6.14.1 Recommended equipment

Role	Recommended	Alternatives / notes	Approx. cost
Mobile scan terminal (floor)	Zebra TC22 — Android, built-in 2D imager, Wi-Fi	Honeywell CT30 XP; any rugged Android terminal whose scanner runs in keyboard-wedge mode	~$1,100
Pilot / backup scanner	Any Android phone using the ▯ camera button	A ~$60 Bluetooth 2D scanner paired to a phone or tablet also works	$0–100
Receiving / office station	Corded USB 2D scanner (e.g. Zebra DS2208) plugged into the station PC	Presents as a keyboard; scans land in whatever field has focus	~$60
Label printer (desktop volume)	Zebra ZD421, 4-inch, **thermal transfer**	Honeywell PC42t. Choose 300 dpi if you will print small labels or dense QR codes; 203 dpi is fine for 4×2-inch and larger	~$500

Role	Recommended	Alternatives / notes	Approx. cost
Label printer (heavy volume)	Zebra ZT231 industrial	Only needed above roughly a few thousand labels/week or in a dusty/hot dock environment	~$1,300
Wi-Fi coverage	Site-survey the warehouse; add access points until there are no dead aisles	Warehouse Mode is online-only — a dead zone is a stopped worker	varies

Buy **one terminal and one printer first**, prove the workflow, then expand. A single 4-inch printer covers every label type below.

1.6.14.2 Choosing print technology and label media

Thermal printers come in two flavors, and the choice is per-label-type, not per-printer (a thermal-transfer printer can do both):

- **Thermal transfer** (ribbon) — the image is fused from a ribbon and does not fade. Required for anything that must survive years, sunlight, or abrasion: **bin/rack/shelf labels**.
- **Direct thermal** (no ribbon) — cheaper per label, but the image fades in months, faster in heat or sun. Fine for anything with a short life: **carton labels, lot labels, pick labels**.

Label type	Media	Size that works well	Technology
Bin / rack / shelf	Polypropylene, permanent adhesive	4×2 in (bins), 4×6 in (rack ends)	Thermal transfer, resin or wax-resin ribbon
Item / carton	Paper, permanent adhesive	4×2 in	Direct thermal
Lot (item + lot + expiry)	Paper	4×3 in	Direct thermal
Pallet placard	Paper or polypropylene	4×6 in	Either
Document barcodes (SO/PO/transfer #)	Printed on the document itself by the office laser printer	—	—

Symbology: use **Code 128** for one-dimensional codes (compact, encodes the full item-number character set) or a **QR code** where labels are scanned at odd angles or from a distance — 2D imagers and the camera scanner read both. Be consistent per label type. Leave a quiet zone (blank margin) of at least 1/8 inch around every barcode, and always print the human-readable text under it — people still need to read labels when a battery dies.

1.6.14.3 Printing labels from OpenERP

Two label runs are built in, each with two outputs — **PDF label sheets** (Avery 5163: Letter sheets of ten 4×2-inch labels, printable on any office printer — works with no thermal printer at all) and **direct thermal printing** (sends ZPL to the configured network printer):

- **Bin labels — Inventory → Warehouses → Bins**: the bins modal has □ **Label Sheet (PDF)** and, when a printer is configured, □ **Send to Label Printer**. Prints one label per active bin: barcode + bin code + warehouse + aisle/rack/shelf.
- **PO carton labels — Purchasing → Orders → select a PO → Labels**: choose *Fixed count per line* (e.g. one per carton) or *One per unit ordered*, then PDF or direct print. Each label carries the item-number barcode, description, and the PO number/vendor. Per-unit runs of any size are fine on the thermal printer (the printer repeats the label); PDF sheets are capped at 1,000 labels per run.

To enable direct printing, set the printer's **IP address, port (9100), and head DPI (203/300)** under **Admin → Setup → Company → Label Printing**. Leave the IP blank to work PDF-only. The test of a correct setup: print one bin label and scan it in Warehouse Mode (§5.13).

1.6.14.4 When each label gets printed

When	What to print	How	Why then
Warehouse setup (once), and whenever bins are added	All bin labels for the warehouse	Warehouses → Bins → Label Sheet / Send to Printer	Bins must be scannable before any scan-driven putaway or counting can start
When a PO is placed **— before the container arrives**	One item label per carton expected, from the PO lines	PO list → Labels	The receiving crew labels cartons as they unload; unlabeled goods never enter the racks
At receiving	Lot labels (and pallet placards if palletized)	Warehouse Mode → Receive: after posting, the confirmation card prints labels for the lots just captured (planned)	Lot identity is only reliably known at the dock
When printing any SO/PO/transfer document	The document number as a barcode on the page		One scan at the floor opens the right transaction

1.6.14.5 Placement and quality best practices

- **Consistent placement.** Bin labels on the same corner of every location (bottom-left of the beam is common), at eye height where possible; upper rack levels get labels angled downward or duplicated at eye level with an up-arrow. Carton labels on the same face/corner of every carton.
- **Scan-test every new batch.** Print one label, scan it in Warehouse Mode, and confirm it resolves to the right record *before* printing hundreds.
- **Never hand-write over a barcode** or wrap tape/stretch-film across one — reprint instead. A glossy overlay can defeat the scanner.
- **Replace, don't patch.** A damaged bin label is a reprint, not a marker touch-up; a label that scans wrong is worse than no label.
- **Keep ribbon and media stocked** and matched (wax-resin ribbon on polypropylene, wax on paper). Order media when the last spare roll goes on the printer.

- **Network the printer.** Put the label printer on the LAN with a static IP and configure it under Admin → Setup → Company → Label Printing. OpenERP sends plain ZPL to TCP port 9100 — no driver installation. Zebra's free ZebraDesigner remains handy for one-off custom labels.

1.7 6. Item Master

1.7.1 6.1 Items

Inventory → Items — the central item catalog.

The list includes a **Vendor** column showing each item's preferred vendor (managed under Item Vendors, §6.11 — the ▯ preferred entry wins). Like Type and Category, it carries an Excel-style ▼ filter (§1.3), so "show me everything we buy from vendor X" — or items with **no** vendor assigned, via **(Blanks)** — is one click.

Key fields: - **Item number** — unique; used on all documents - **Description** - **Category** — for reporting and matrix pricing - **UOM** — stocking unit of measure - **Standard cost** — used for GL posting when FIFO layers are unavailable - **List price** — base selling price before price lists apply - **Reorder point / reorder qty / min qty / max qty** — drive reorder suggestions - **ABC class** — A, B, or C (updated by ABC classification run) - **HTS code, country of origin, tariff rate** — for import cost calculations - **Is kit** — if checked, the item is exploded into BOM components on SO entry - **Drawing #, revision, material spec, manufacturing method** — the part's engineering identity (see §6.16); shown as a Drawing # column in the list - **Specifications / critical characteristics** — free-text engineering notes captured at creation or edited later via the **Eng** button

1.7.2 6.2 Categories and UOM

- **Inventory → Categories** — organize items for reporting and matrix pricing
- **Inventory → UOM** — units of measure; add **Conversions** (e.g. 12 EA = 1 CASE) so items can be sold in one UOM and stocked in another

1.7.3 6.3 Price Lists

Sales → Price Lists — customer-assignable price lists with per-item unit price overrides.

- **Items** button on a price list row — add or edit item overrides (unit price, effective/expiry dates)
- **Q Breaks** button on a price list item — define quantity break tiers: `min_qty, unit_price` pairs. When a line's qty_ordered meets a break, the break price is used.

  Pricing hierarchy (highest to lowest priority): 1. Contract price (customer + item + date range) 2. Price list assigned to the customer (with quantity breaks) 3. Matrix pricing (customer class × item category discount) 4. Item list price

  If none of these apply, the order entry user must enter the price manually. Review and update price lists at least annually — or when your cost increases — to protect margins.

1.7.4 6.4 Contract Pricing

Sales → Contracts — customer + item fixed price with an effective date range. Takes priority over all price lists. Good for long-term supply agreements.

1.7.5 6.5 Matrix Pricing

Sales → Matrix Pricing — discount rules keyed on customer class × item category. Applied when no contract or price list override exists. Example: KEY customers get 15% off FITTINGS.

1.7.6 6.6 Lot Tracking

Items with `lot_tracking = true` require a lot number on receipt and shipment. Manage lot numbers from **Inventory → Items → Lots** modal: enter lot number, received date, and optional expiry date. Expiring lots are highlighted in the lot list.

1.7.7 6.7 Serial Number Tracking

Items with `serial_tracking = true` require a serial number on each unit shipped. Manage serial numbers from **Inventory → Items → Serials** modal: enter serial number, received date, optional warranty expiry.

1.7.8 6.8 Traceability

Inventory → Traceability — look up all inventory transactions for a given serial number or lot number. Trace from receipt PO through shipment SO.

1.7.9 6.9 Item Cross-References

Inventory → Items → X-Refs modal — store alternative part numbers:

Type	Example use
CUSTOMER	Customer's own part number for this item
COMPETITOR	Competitor's catalog number
UPC	Universal Product Code
MPN	Manufacturer Part Number

Cross-references feed the barcode lookup endpoint.

1.7.10 6.10 Item Substitutes

Inventory → Items → Subs modal — define substitute items with priority ranking. When a flagged-superseded item is selected on an SO, the system prompts with available substitutes.

1.7.11 6.11 Item Vendors

Inventory → Items → Vendors modal — record which vendors supply this part. Most parts can be bought from more than one supplier, so this isn't a single "vendor" field on the item — it's a list: pick a vendor from the dropdown, then capture that vendor's own SKU for this item, lead time, minimum order quantity (MOQ), packing quantity, and last cost. Add as many vendors as actually supply the part, and star one as **preferred**.

This is the same underlying data as the Vendor Item Catalog (§4.11) — the two screens are just two ways of looking at it, and either one can add, edit, or remove a link: - **Inventory → Items → Vendors** answers "which vendors supply *this* part" — you start from the item. - **Purchasing → Vendor Catalog** answers "everything *this* vendor supplies" — you start from the vendor.

Use whichever direction matches the question you're actually asking; changes made from one show up in the other immediately, since nothing is duplicated behind the scenes.

1.7.12 6.12 Item Supersession

Inventory → Items → Supersession history — record when one item replaces another (old item → new item, reason, effective date). SO entry warns the user when a superseded item is selected.

1.7.13 6.13 Kitting / BOM

Inventory → Kitting — the bill of materials for kit items: which component items go into a kit, and how many of each. When a kit is shipped, the system pulls FIFO cost from its components rather than from the kit itself.

Building a BOM: 1. Pick the kit item from the dropdown at the top. (Kits are items flagged as such; use **Show all items** if the one you want isn't listed.) 2. If it has no BOM yet, click **Create BOM**. 3. Click **+ Add Component** to add each component item with its quantity per kit, unit of measure, and an optional-component flag. 4. **Qty per Kit is editable in place** — click the cell in the grid, type the new number, and press Enter. It saves immediately, so correcting a quantity doesn't need the Add dialog. 5. To remove a component, select its row and click □ **Remove** in the action bar.

Explode Kit… previews what building a given quantity of the kit would consume: each component, the quantity needed, what's on hand, and whether you're short. Run it before promising a build date — it's the difference between "we can ship 50" and "we can ship 50 except for the seals."

1.7.14 6.14 Barcode Scanning

The endpoint `GET /items/barcode/{code}` checks `item_number` first, then cross-reference numbers (UPC, MPN, etc.). Returns the matching item. Useful for integrating mobile scan devices or the WH app.

1.7.15 6.15 Inbound Shipments

Purchasing → Inbound Shipments — track ocean or air freight containers with vessel, voyage, carrier, origin/destination ports, ETA, ATA, and customs status (PENDING → IN_TRANSIT → ARRIVED → CLEARED → HELD).

Attach POs to a shipment to group expected receipts by container.

1.7.16 6.16 Engineering Data and Part Documents

For custom-engineered parts, an item is defined by its drawing, not just its number. Select an item and click **Eng** in the action bar to open the Engineering panel:

- **Engineering fields** — drawing #, current revision, material spec, manufacturing method, plus free-text specifications and critical characteristics. Editable here at any time (the New Item form also captures them at creation). The list grid shows a **Drawing #** column with the current rev.
- **Part documents** — upload drawings, spec sheets, CAD models, material certs, quality docs, and photos (max 25 MB each). Each file can carry its own **Rev** tag, so when the drawing moves from rev C to rev D you keep both PDFs and can tell them apart. Removing a document deactivates it; nothing is destroyed.

> **Inquiry Master hand-off:** when an inquiry is awarded (§3.10) and a new item is created, the drawing #, revision, and material spec entered on the inquiry line are copied onto the item automatically — the engineering identity established during quoting follows the part into production without re-typing.

1.8 7. Financials

The Financials module provides the General Ledger, financial statements, bank reconciliation, multi-currency, and year-end close. For most companies, the subledgers (AR, AP, Inventory) handle the day-to-day posting automatically. The GL is where you review the cumulative result, make manual corrections, and produce financial statements for management and external reporting.

1.8.1 7.1 General Ledger

GL → Accounts — chart of accounts (see §1.5).

GL → Fiscal Years / Periods — set up the accounting calendar. Closing a period prevents any further posting to it. Fiscal year close (§7.7) is a separate workflow.

GL → Journals — manual journal entry:

1. Create a batch (batch number, description, period).
2. Add a journal entry (entry number, date, source, reference, description).
3. Add debit and credit lines. The batch will not post unless total debits = total credits.
4. **Post** the batch. Once posted, entries cannot be edited; a reversing entry is required.

> **When to use manual journals:** - Depreciation not handled by the Fixed Assets module - Accruals (record an expense incurred but not yet billed) - Reversals (reverse a prior period accrual) - Reclassifications (move an amount from one account to another) - Opening balance entries at go-live
>
> Do **not** use manual journals for: AR invoices, AR payments, AP bills, AP payments, SO shipments, or inventory adjustments. Those events post automatically. Creating a manual journal for something the system already posted will double-count the amount.

Subledger-to-GL auto-posting happens silently when AR invoices, AR payments, AP bills, AP payments, SO shipments, and inventory adjustments are saved. You generally do not need to create manual journals for those events.

1.8.2 7.2 Financial Statements

GL → Reports: - **Trial Balance** — all accounts with period debit/credit activity and ending balance. Requires a period selection. - **Income Statement** — revenue minus expense for a period or date range. - **Balance Sheet** — assets, liabilities, and equity as of a period end.

> **How to read the Trial Balance as a diagnostic:** The Trial Balance should always balance (total debits = total credits). Run it at month end before closing the period. Look for: - Accounts with balances in the wrong direction (e.g., a REVENUE account with a debit balance — may indicate a void posted to the wrong period) - Unexpectedly large or small balances compared to prior months - Accounts with activity you do not recognize

> Investigate anomalies before closing the period. It is much harder to correct after the period is closed.

1.8.3 7.3 GL Budgets

GL → Budgets — enter budget amounts per account per period. The **Budget vs Actual** report compares YTD actual to YTD budget.

1.8.4 7.4 Departments

GL → Departments — cost centers (e.g. SALES, WAREHOUSE, ADMIN). Journal lines can be assigned to a department for departmental P&L analysis.

1.8.5 7.5 Fixed Assets and Depreciation

GL → Fixed Assets — track capital assets with: - In-service date, useful life (months), salvage value - Linked GL asset account and accumulated depreciation account - **Depreciate** button — posts straight-line monthly depreciation entries for the selected period

GL → Depreciation Report — shows all assets with book value, accumulated depreciation, and remaining life.

> **Monthly depreciation procedure:** At month end, go to **GL → Fixed Assets** and click **Depreciate** for each active asset. This posts the depreciation journal entry for that period. Do this before running the Income Statement so that the depreciation expense is included in the current period's result.

1.8.6 7.6 Recurring Journal Entries

GL → Recurring JEs — templates for monthly entries (rent, amortization, etc.): - Define template lines (debit/credit amounts and accounts). - Set frequency (MONTHLY, QUARTERLY, ANNUALLY) and next run date. - **Run** generates a journal batch from the template and advances the next run date.

> **Common recurring entries to set up:** - Rent (DR Rent Expense / CR Prepaid Rent or CR Accounts Payable) - Insurance prepaid amortization (DR Insurance Expense / CR Prepaid Insurance) - Loan interest accrual (DR Interest Expense / CR Accrued Interest) - Deferred revenue recognition (DR Deferred Revenue / CR Revenue)
>
> Setting these up as recurring templates ensures they are never forgotten and the amounts are consistent each month.

1.8.7 7.7 Year-End Close

GL → Year-End Close (when fiscal year is fully posted): 1. Close all open periods in the fiscal year. 2. **Run Close** — the system: - Zeros all revenue and expense account balances for the year via a closing journal entry - Credits the net income to the Retained Earnings account - Marks the fiscal year CLOSED 3. Create the new fiscal year and its periods.

> **Year-end close is irreversible.** Before running it, confirm: - All AR invoices and payments for the year are entered - All AP bills and payments for the year are entered - All physical count variances are posted - All fixed asset depreciation entries are posted - All recurring journal entries have been run for December - The Trial Balance balances and all balances look correct - A backup of the database has been taken

> The year-end close does not affect AR or AP balances — those carry forward as open items. Only income statement accounts (Revenue, Expense) are zeroed.

1.8.8 7.8 Bank Reconciliation

Financials → Bank Reconciliation: 1. Select a bank account and date range. 2. **Add Statement Lines** — enter or import bank statement transactions (deposits positive, payments negative). 3. **Match** — link each statement line to an AR receipt (deposit) or AP payment from the system. 4. Summary tiles show: statement total, cleared total, uncleared total, outstanding deposits, outstanding payments. 5. The reconciliation is complete when uncleared items equal the difference between the bank statement balance and the GL cash account balance.

> **Bank reconciliation procedure (monthly):** 1. Obtain the bank statement (paper, PDF, or CSV download from your bank's portal). 2. Enter all statement lines into the system using **Add Statement Lines**, or import them. 3. Match each statement line to a payment or deposit in the system using the **Match** button. 4. Common unmatched items: - **Outstanding checks** — payments created in the system but not yet cleared at the bank - **Deposits in transit** — AR receipts recorded in the system but not yet on the statement - **Bank charges** — service fees on the statement with no corresponding system entry; create a manual journal (DR Bank Charges / CR Cash) 5. The reconciliation is complete when: **Bank Statement Balance + Outstanding Deposits − Outstanding Checks = GL Cash Account Balance**. 6. Reconcile monthly, ideally within 5 business days of receiving the statement.

1.8.9 7.9 Multi-Currency

Admin → Currencies — activate currencies and set exchange rates (spot rates by date).

Foreign-currency AP bills store the foreign amount and the exchange rate at bill date. When the bill is paid at a different rate, the system computes and posts FX gain/loss to GL automatically.

GL → FX Revaluation — revalue all open foreign-currency AR/AP balances at the current exchange rate and post unrealized gain/loss.

> **Multi-currency procedure:** At month end, if you have open foreign-currency AR or AP balances, run **FX Revaluation** to record unrealized gains and losses at the current month-end rate. This is required for accurate financial statements under US GAAP and IFRS. The revaluation entries reverse in the following period, so only the realized gain/loss (from payment) remains permanently.

1.8.10 7.10 Cash Flow Statement

GL → Cash Flow — indirect method: starts from net income, adds back depreciation, adjusts for AR/AP/inventory working capital changes, and includes asset purchases. Reads from GL account balances by subtype.

1.8.11 7.11 Cash Application Batch

AR → Batch Apply — apply multiple payments at once. The system matches each payment to the customer's oldest open invoices first (oldest-invoice-first auto-apply). Returns a per-payment result showing applied amounts and any unapplied balance.

1.9 8. Administration

1.9.1 8.1 Users and Groups

Admin → Users: - Create users with a username, password (minimum 6 characters), full name, and **one or more groups** — check every group that applies in the scrollable Groups checklist. At least one group is required. - The Users list shows all of a user's assigned groups, comma-separated, in the **Groups** column. - **Email** is optional but recommended for every user — it's validated for a proper email format and is stored for the user's account, ready for future features (e.g., reminders/notifications) that need somewhere to send them. It is not currently used to send anything. - Deactivate users to block login without deleting records. - **Change Password** — bumps `token_version` to immediately invalidate all active sessions for that user. - **⊘ Force Logout** — invalidates all active tokens for the user without changing the password.

Admin → Groups: - Groups bundle permissions. A user's effective permissions are the **union** of every group they're assigned to — checking a second group only ever adds access, it never takes any away. - Create groups with a name and description, then check the permissions to grant. - Changing a group's permissions takes effect on the user's next request (token is re-validated per request); adding or removing a group from a user takes effect the next time that user logs in.

> **Recommended group structure:**
>
> | Group | Who | Typical permissions |
> |---|---|---|
> | ADMIN | Controller, IT | All permissions |
> | AR_STAFF | AR clerk | AR module only |
> | AP_STAFF | AP clerk | AP module only |
> | WAREHOUSE | Receiving, shipping | Inventory, PO receive |
> | SALES | Inside sales | SO, quotes, customers |
> | READ_ONLY | Managers, auditors | View only |
>
> Separate AR and AP access so that no single employee can both record a payment receipt AND apply it to an invoice without a second person reviewing. This is a basic internal control.
>
> For staff who genuinely cover more than one role (e.g. an office manager who does both AR and light warehouse work), assign both groups directly to that one user rather than creating a new combined group — it keeps the group list itself small and each group's purpose single-purpose and auditable.

1.9.2 8.2 Permissions Reference

Permission key	What it gates
`system.users.manage`	Admin → Users and Admin → Groups
`system.config.manage`	Admin → Company Settings, Audit Log
`gl.reports.view`	GL Trial Balance, Income Statement, Balance Sheet
`ar.invoices.view`	AR → Recurring Invoice Templates
(all other modules)	`require_auth()` only — any logged-in user can access

Note: most module routes require only a valid login (not a specific permission). Per-route permission enforcement is an active roadmap item (D4).

1.9.3 8.3 Audit Log

Admin → Audit Log — every data-changing API call logs: table name, record ID, action (CREATE, UPDATE, DELETE, POST, APPLY, SHIP, etc.), changed-by username, and timestamp. Filter by table, user, or date range above the grid.

Seeing exactly what changed. Select any entry in the grid and a **Before / After** panel opens underneath it, showing the record's field values as they were and as they became. This is the heart of an investigation: it's not just *that* someone changed a credit limit, it's that they changed it from 5,000 to 50,000 at 4:58 PM on a Friday.

> **Using the audit log for investigations:** If a balance is unexpectedly wrong, filter by the affected table and date range, then click through the entries for that record and read the Before/After panel on each. You can see exactly who changed what, when, and from what to what. The audit log is append-only and cannot be edited.

1.9.4 8.4 Payment Terms

Admin → Payment Terms — manage NET30, NET60, 2/10-NET30, etc. Payment terms drive due-date calculation on invoices and bills.

1.9.5 8.5 Tax Codes

Admin → Tax Codes — simple flat-rate tax codes (e.g. TX-STATE at 6.25%). Assigned to items; defaulted onto invoice lines.

1.9.6 8.6 Sales Tax (US Multi-Jurisdiction)

Admin → Sales Tax — jurisdiction-based rates: - Create jurisdictions at state, county, and city level with state/county/city rate components. - Assign **exemption certificates** to customers (expiry date, certificate number) — exempted customers have 0% tax applied. - On an AR invoice, the **Sales Tax** panel computes the blended rate for each line's jurisdiction and applies it via **Apply to Lines**.

1.9.7 8.7 Customer Classes

Admin → Customer Classes — codes used for matrix pricing and reporting (e.g. KEY, STANDARD, PROSPECT). Assign to customers on the customer form.

1.9.8 8.8 Return Reason Codes

Admin → Return Reasons — standardized codes for RMA lines (e.g. DEFECTIVE, WRONG_ITEM, DAMAGED). Appear in the RMA line form dropdown.

1.9.9 8.9 Carriers and Ship Methods

Admin → Carriers — carrier master (UPS, FedEx, etc.). Linked to ship methods.

Admin → Ship Methods — specific service levels per carrier (UPS GROUND, FEDEX 2DAY). Assignable on SOs and POs.

1.9.10 8.10 SMTP and Email Log

Configure SMTP under **Admin → Company Settings** (smtp_host, smtp_port, smtp_encryption, smtp_user, smtp_pass, smtp_from). Use **Test Send** to verify delivery.

Admin → Email Log — every email sent by the system (invoice, SO, PO) is recorded with status, recipient, subject, and timestamp.

1.9.11 8.11 Scheduled Reports

Yes — these reports auto-email, to whoever you list as recipients when you set up the schedule. There is no automatic distribution list or "send to all users" option: you type in the exact email addresses that should receive each report, and only those addresses get it.

1.9.11.1 How to set who gets what report

Go to **Admin → Scheduled Reports → + New Schedule** and fill in:

Field	What it controls
Report Name	A label for you to recognize this schedule by (e.g. "Weekly AR Aging for CFO")
Report Type	One of: AR Aging, AP Aging, Trial Balance, Back-Orders, Commissions
Frequency	Daily, Weekly, Bi-Weekly, Monthly, or Quarterly
Next Run Date	The first date this should fire. After each send, this auto-advances by the frequency (e.g. a Weekly report set to run 2026-07-06 becomes due again on 2026-07-13)
Format	**CSV** (a plain data table, shown inside the email body) or **HTML** (a formatted table inside the email body) — either way, the report is embedded directly in the email itself, not sent as a file attachment
Recipients	**This is the "who gets it" field.** Comma-separated email addresses — e.g. `cfo@company.com`, `controller@company.com`. Every address here receives every run of this specific schedule
Active	Turn a schedule off without deleting it (useful when someone's on leave, or a report is temporarily not needed)

Each report type pulls whatever data matches its name at the moment it runs (e.g. AR Aging always reflects that day's open invoices) — you aren't choosing which customers/vendors/accounts, just which report and who receives it. Want two different groups to get different slices (e.g. Trial Balance to Finance, AR Aging to Sales)? Create two separate schedules — one per report type/recipient combination.

1.9.11.2 How the automatic emailing actually happens

Two things both need to be true for a report to actually land in someone's inbox:

1. **SMTP must be configured** under **Admin → Company Settings → Email / SMTP** (host, port, from address, credentials). If this is missing, the runner still generates the report and marks the schedule as run, but the send is silently skipped — check **Admin → Email Log** (§8.10) to confirm whether emails are actually going out, and to see the exact error if a send fails.
2. **Something has to trigger the check.** OpenERP does not run reports on its own in the background — a separate command runs once, checks which schedules are due, sends them, and exits:

   ```
   C:\php\php.exe -c C:\php\php_sapads.ini C:\OpenERP\tools\run_scheduled_reports.php
   ```

   This command only sends reports whose **Next Run Date** is today or earlier — running it twice in the same day does not double-send anything already sent, since each successful run pushes Next Run Date forward.

1.9.11.3 Making it actually automatic

The command above does nothing by itself unless something runs it on a schedule. The recommended way is the **OpenERP Worker Service** (`service\OpenERPSvc.exe` — see `service\README.md`): a small Windows service that already ships configured to run this script daily at 6:30 AM (plus the message-queue worker every minute and the recurring-PO runner daily). Install it once as Administrator:

```
cd <install dir>\service
.\OpenERPSvc.exe I
sc start OpenERPSvc
```

It auto-starts with Windows, restarts itself after a crash, logs every run to `service\OpenERPSvc.log`, and its schedule is edited in plain `service\OpenERPSvc.ini` (changes take effect without a restart).

If you prefer not to install the service, classic **Windows Task Scheduler** works too:

1. Open **Task Scheduler** (search for it in the Start menu).
2. **Create Task...** (not “Basic Task” — you want more control over timing).
3. **General tab:** name it something like “OpenERP Scheduled Reports”; select “Run whether user is logged on or not.”
4. **Triggers tab → New:** Daily, start time e.g. 7:00 AM (run it once a day even if you have weekly/monthly schedules — the script only sends what’s actually due that day).
5. **Actions tab → New:**
   - **Program/script:** `C:\php\php.exe`
   - **Add arguments:** `-c C:\php\php_sapads.ini C:\OpenERP\tools\run_scheduled_reports.php`
6. **OK**, entering the credentials of an account with permission to run scheduled tasks on the server.

From then on, Windows itself fires this script daily; OpenERP checks what’s due and emails it — no one has to remember to run it by hand.

1.9.12 8.12 CSV Import

Admin → Import Tool — bulk-load data instead of hand-keying it. There are two situations where this is worth doing rather than typing records in one at a time:

- **Go-live conversion.** A new company coming onto OpenERP almost always already has this data somewhere — an old accounting system, a spreadsheet, an export from the previous ERP. Re-typing a chart of accounts, a customer list, or an item catalog by hand is slow and error-prone (a mistyped credit limit or GL account number is a silent, expensive mistake). Importing it is faster and lets you validate the data once, in bulk, instead of trusting hundreds of individual manual entries.

- **Bulk additions to a live system.** Even after go-live, data keeps arriving in bulk from outside the system — a vendor sends a 500-line price list, a new warehouse needs its bin layout loaded, a sales rep hands over a spreadsheet of open quotes to convert. The Import Tool handles these the same way, and is always safe to run against a system that already has data in it (see the rule below).

Every import type shares the same rule:

> **A row whose natural key already exists is left untouched and reported as "skipped — already exists." Imports never overwrite existing records.** It is always safe to re-run the same file (e.g. after fixing a few rows) — anything already loaded is simply skipped the second time.

1.9.12.1 8.12.1 Preparing Your Data in Excel

The Import Tool takes one CSV file per import type, but there's nothing wrong with building all of your source data in a single Excel workbook first — it's the easiest way to keep everything organized while you assemble it, cross-check codes between tabs, and fix mistakes before anything touches the live system.

1. **One tab per import type**, named to match the Import Tool's dropdown (e.g. `Customers`, `Chart of Accounts`, `Sales Orders`). This keeps the workbook organized and makes it obvious which tab feeds which import.
2. **Row 1 of every tab is the header row**, with column names copied exactly from the Import Tool page for that import type (letter case doesn't matter — the importer lower-cases everything — but spelling and underscores must match, e.g. `customer_number` not `Customer Number`). Use the **Download Template CSV** button on the Import Tool page to get a ready-made header row for any import type; paste it into row 1 of the matching tab.
3. **One row per record** on flat sheets (Customers, Vendors, Items, Chart of Accounts, Categories, UOM, Warehouses, Bins, Price Lists, Price List Items, Vendor Catalog, GL Opening Balances). **One row per line item** on the four multi-line document sheets (Sales Orders, Purchase Orders, AR Invoices, AP Bills) — see §8.12.3.

Formatting gotchas that cause otherwise-correct data to fail on import:

Problem	What happens	Fix
Dates in a regional/auto format	Excel may store a different date than what's displayed, and the CSV holds the stored value	Format date columns as plain **Text**, or type dates as `YYYY-MM-DD` (e.g. `2026-07-01`)
Codes that look numeric (`account_number`, `zip`, `uom_code`, item numbers like `6E10`)	Excel drops leading zeros or converts to scientific notation	Format the column as **Text** *before* typing values into it
Percentages (`discount_pct`, tax rates)	A cell formatted as "5%" is stored as `0.05`, not `5`	Format the column as a plain **Number**, and type the percent value itself (`5` for 5%)
Trailing/leading spaces on codes used for lookups (`warehouse_code`, `terms_code`)	`"MAIN"` and `"MAIN "` look identical to a person but fail to match in the importer	Use Excel's `TRIM()` function on a helper column, then paste the result back as values

Problem	What happens	Fix
Non-ASCII characters (accented names, curly quotes pasted from Word)	Can get mangled if saved with the wrong CSV encoding	Save as **CSV UTF-8**, not plain **CSV** (see below)

Exporting a tab to CSV: CSV is a single-sheet format — Excel cannot export a whole workbook to one CSV, only the *active* tab. To export each tab you plan to import:

1. Click the tab you want to export so it's the active sheet.
2. **File → Save As** → choose **CSV UTF-8 (Comma delimited) (*.csv)** as the file type (not plain "CSV" — UTF-8 avoids mangled characters).
3. Give it a name you'll recognize (e.g. `customers.csv`, `sales-orders.csv`) and save.
4. Excel will warn that "the selected file type does not support workbooks with multiple sheets" — click **OK**; this only means it's exporting the current tab, which is exactly what you want.
5. Repeat once per tab. Keep your master workbook (with all tabs) as the source of truth — re-export a fresh CSV from it any time you need to fix and re-run an import.

1.9.12.2 8.12.2 What Each Import Type Is For

Category	Import type	Sheet contents	Why import it	Natural key
Master Data	Customers	One row per customer: name, address, credit limit, terms, tax exemption	The core of every sale — load your existing customer list instead of re-typing it during onboarding	`customer_number`
	Vendors	One row per vendor: name, address, tax ID, 1099 flag, terms	Needed before you can enter POs or bills; brings your existing vendor list over as-is	`vendor_number`
	Vendor Contacts	One row per person: vendor, name, title, department, email/phone/WhatsApp/WeChat	Loads each vendor's department contacts (Sales, Quality, Engineering…) in bulk instead of adding them one at a time on the Contacts tab	`vendor_number` + `contact_name`

Category	Import type	Sheet contents	Why import it	Natural key
	Items	One row per item/SKU: description, category, UOM, cost, price	Usually the largest single import for a new company — a catalog of hundreds or thousands of SKUs is impractical to key by hand	`item_number`
Setup & Lookups	Chart of Accounts	One row per GL account: number, name, type, normal balance, parent	The financial backbone of the system — import it once from your existing books rather than re-building it by hand	`account_number`
	Item Categories	One row per category, optionally nested via `parent_code`	Only useful if items are imported afterward and reference these codes; lets you preserve an existing category hierarchy	`category_code`
	Units of Measure	One row per UOM (EA, CS, BX, etc.)	A short list, but items reference it — load it before Items if you use non-default UOM codes	`uom_code`
	Warehouses	One row per warehouse: code, name, address	Needed before importing bins or any inventory-aware transaction	`warehouse_code`
	Bin Locations	One row per bin: warehouse, bin code, aisle/rack/shelf	Lets you bring in an existing rack layout instead of creating bins one at a time in the UI	`warehouse_code` + `bin_code`

Category	Import type	Sheet contents	Why import it	Natural key
	Price Lists	One row per price list (RETAIL, WHOLESALE, etc.)	Needed before Price List Items	`price_list_code`
	Price List Items	One row per item price on a list	Brings in existing customer/tier pricing instead of re-keying every price	`price_list_code` + `item_number`
	Vendor Item Catalog	One row per vendor/item pairing: vendor SKU, lead time, MOQ, cost	Common in purchasing-heavy conversions where a vendor's SKU differs from your own	`vendor_number` + `item_number`
Opening Balances	GL Opening Balances	One row per account/period: debit or credit amount	The standard way to convert an accounting system — load the trial balance as of cutover instead of replaying every historical journal entry. **Do this last** — see §8.12.4	`account_number` + `period_code`
Transactions	Sales Orders	One row per SO line, grouped by `so_number`	For orders that are still open when you cut over — they need to exist in OpenERP so they can be shipped/invoiced going forward	`so_number`
	Purchase Orders	One row per PO line, grouped by `po_number`	Same idea for open purchase orders still awaiting receipt at cutover	`po_number`

Category	Import type	Sheet contents	Why import it	Natural key
	AR Invoices	One row per invoice line, grouped by `invoice_number`	Brings in *open* customer invoices so collections/aging work correctly after cutover, without re-entering each one by hand	`invoice_number`
	AP Bills	One row per bill line, grouped by `bill_number`	Same idea for open vendor bills still awaiting payment	`bill_number`

Required and optional columns for the selected import type are always listed on the Import Tool page itself, and in the header row of its downloadable template.

1.9.12.3 8.12.3 Multi-line Document Sheets

Sales Orders, Purchase Orders, AR Invoices, and AP Bills are **multi-line documents** — the CSV has **one row per line item**, not one row per document. Rows that share the same document number (`so_number`, `po_number`, `invoice_number`, or `bill_number`) are grouped into a single document; header fields (customer/vendor, dates, terms) are read only from the **first** row of each group, so repeat them on every line for clarity but know that only the first is used. If any line in a document fails validation, the whole document is rejected — nothing is partially imported. Sales Orders have no credit-hold/credit-limit check on import (that check applies when a user creates an order by hand).

1.9.12.4 8.12.4 Recommended Order for a Full Conversion

When loading an entire new company's data, import in this order — each step depends on data from the one before it:

1. **Chart of Accounts** (and Payment Terms / Tax Codes, entered manually under Admin — see §1.6)
2. **Item Categories** and **Units of Measure**
3. **Warehouses**, then **Bin Locations**
4. **Customers** and **Vendors**
5. **Items**
6. **Price Lists**, then **Price List Items**; **Vendor Item Catalog** if used
7. Open **Sales Orders / Purchase Orders** still in flight at cutover
8. Open **AR Invoices / AP Bills** still outstanding at cutover
9. **GL Opening Balances**, sized to include the AR/AP control account balances that step 8 just created

Why GL Opening Balances goes last, and why AR Invoices/AP Bills don't post to GL on import: creating an invoice or bill by hand in the app automatically posts it to the General Ledger (DR AR/CR Revenue, DR Expense/CR AP). Imported invoices/bills deliberately skip that step, because the opening trial balance in step 9 is loaded as a single lump sum per account — it already reflects the net effect of every open invoice and bill from step 8. Posting them individually on top of that lump sum would double-count the AR/AP and revenue/expense balances. This was a deliberate design choice for the conversion scenario; it means an import can never be used to feed GL-posted transactions into an already-live, already-balanced ledger — only manual entry (or the SO ship / PO receive / payment-apply flows) posts to GL after go-live.

1.9.12.5 8.12.5 Reading the Import Report

After each import, the page shows tiles for Total / Imported / Skipped / Failed, followed by a detail table (one row per document for multi-line types, one row per CSV row otherwise) with the natural key, a status badge, and a message explaining *why* — e.g. "Customer number CUST-1002 already exists" (skipped) or "Item ITM-500 not found" (failed). Check "Hide successfully imported rows" off to review the full file, including the successes.

1.9.12.6 8.12.6 Preparing the Vendor Master Spreadsheet

The Vendors import loads the full Vendor Master (§4.1) — agents, manufacturers, and service providers — from a single spreadsheet. Build one Excel tab named `Vendors`, **one row per company** (an agent is a row; each of its factories is its own row), with the columns below in row 1. Only the first two columns are required; leave any cell blank to accept the default or skip the field. The **Download Template CSV** button on the Import Tool page produces this exact header row.

Identity (required):

Column	What to enter
`vendor_number`	Your unique code for the company, e.g. `AGT-WENZHOU` or `MFG-RUIAN`. This is the key used everywhere else — pick a scheme you can live with
`company_name`	Legal business name

Classification and the agent relationship:

Column	What to enter
`vendor_type`	`AGENT`, `MANUFACTURER` (default if blank), `TRADING_COMPANY`, `LOGISTICS`, `INSPECTION`, `WAREHOUSE_3PL`, `CUSTOMS_BROKER`, or `UNKNOWN_MFG`
`status`	`APPROVED` (default), `CONDITIONAL`, `PROBATION`, `INACTIVE`, or `BLOCKED`
`agent_vendor_number`	On a manufacturer row: the `vendor_number` of its agent. The agent's own row can be anywhere in the same file — order doesn't matter. Leave blank for direct manufacturers and for the agents themselves

Company profile:

Column	What to enter
`trade_name`	Trading/brand name if different from the legal name
`address1`, `address2`, `city`, `state`, `zip`	Street address
`country`	2-letter code: `CN`, `TW`, `US`, …
`phone`, `email`, `website`	Main company channels
`contact_name`	The main contact person (department contacts — Sales, Quality, Engineering… — are loaded afterward with the Vendor Contacts import, §8.12.7, or added on the vendor's Contacts tab)
`incoterms`	Default shipping terms: `FOB`, `CIF`, `EXW`, `DDP`, …
`year_established`	e.g. `1998`
`employee_count`	Approximate headcount
`lead_time_notes`	Free text, e.g. `45-60 days after PO`
`company_description`	What they make / specialties
`internal_notes`	Your private notes — never visible to the vendor

Financial:

Column	What to enter
`tax_id`	EIN/SSN — needed for 1099 vendors
`is_1099`	`1` = yes, `0` = no (default)
`is_active`	`1` = active (default), `0` = inactive
`terms_code`	Must match a Terms Code under Admin → Payment Terms, e.g. `NET30`

Factory profile (manufacturer rows only — filling any of these creates the factory profile automatically):

Column	What to enter
`factory_code`	Your short code for the plant, e.g. `RPC-1`
`province_state`	e.g. `Zhejiang`
`time_zone`	e.g. `CST +8`
`primary_language`	e.g. `Mandarin`
`plant_manager`, `quality_manager`, `production_manager`	Names of the key people
`capacity_notes`	Lines, monthly tonnage, max part size…
`equipment_notes`	CNC count, presses, furnaces, CMM…
`quality_notes`	ISO/API certifications held, inspection and lab capabilities

A minimal example — one agent with two factories (one still unidentified) and one direct manufacturer:

```
vendor_number,company_name,vendor_type,agent_vendor_number,country,city,incoterms
AGT-WENZHOU,Wenzhou Sourcing Partners,AGENT,,CN,Wenzhou,FOB
MFG-RUIAN,Ruian Precision Casting Co,MANUFACTURER,AGT-WENZHOU,CN,Ruian,
UNK-WENZHOU,Unknown Manufacturer (Wenzhou SP),UNKNOWN_MFG,AGT-WENZHOU,CN,,
MFG-NINGBO,Ningbo Dongxin Valve,MANUFACTURER,,CN,Ningbo,FOB
```

> **Tips:** Keep `vendor_type`, `status`, and `agent_vendor_number` spelled exactly as shown (case doesn't matter, but `MANUFACTURE` or a typo'd agent number won't match — a bad agent number shows as a *failed* "(agent link)" row in the import report while the vendor itself is still created, so you can fix the link on the vendor's edit form). Format the `vendor_number` and `zip` columns as **Text** in Excel before typing, per the gotchas in §8.12.1. Re-running the same file is safe: existing vendor numbers are skipped, never overwritten.

1.9.12.7 8.12.7 Preparing the Vendor Contacts Spreadsheet

The **Vendor Contacts** import bulk-loads each vendor's people — the department contacts shown on the vendor's Contacts tab (§4.1). Import Vendors first; build one sheet with **one row per person**:

Column	What to enter
`vendor_number` *(required)*	The vendor the person belongs to — must already exist
`contact_name` *(required)*	Full name. A vendor's existing contact with the same name is skipped, never overwritten
`title`	e.g. `Sales Manager`
`department`	`SALES`, `ENGINEERING`, `QUALITY`, `PRODUCTION`, `LOGISTICS`, `ACCOUNTING`, or `MANAGEMENT`
`email`, `phone`, `mobile`	Contact channels
`whatsapp`, `wechat`	Common for overseas vendors — WhatsApp number / WeChat ID
`preferred_method`	`EMAIL`, `PHONE`, `WHATSAPP`, or `WECHAT`
`is_primary`	1 marks this person as the vendor's primary contact (clears any prior primary)
`notes`	Free text

Example:

```
vendor_number,contact_name,title,department,email,preferred_method,is_primary
AGT-WENZHOU,Li Wei,Sales Manager,SALES,liwei@wzsourcing.cn,WECHAT,1
AGT-WENZHOU,Chen Fang,QA Coordinator,QUALITY,chen@wzsourcing.cn,EMAIL,0
MFG-NINGBO,Zhang Ming,Plant Manager,MANAGEMENT,zhang@dongxin.cn,WHATSAPP,1
```

The natural key is `vendor_number` + `contact_name`, so re-running the same file is safe — rows already loaded are reported as *skipped*. An unknown `vendor_number` fails just that row; the rest of the file still imports.

1.9.13 8.13 CSV Export

Most list screens have an **Export CSV** button that downloads the current filtered/paginated view as a CSV file. Available on: AR Invoices, AR Customers, AP Vendors, AP Bills, Sales Orders, and the Inventory Valuation report.

1.9.14 8.14 AI Help Assistant

Every page can show a floating □ **AI Help** button (bottom-right corner). Clicking it opens a chat panel where any user can ask questions in plain English — *"How do I receive a purchase order?"*, *"What does the BLOCKED vendor status do?"*, *"How do I post a physical count?"* — and get an answer drawn from this User Guide, with the relevant section number cited.

What it knows. The assistant answers from two sources that are sent along with every question: this User Guide, and an auto-generated *schema digest* describing the structure of the database (table and column names with their meanings — never any data). If a question isn't covered by the guide, it says so rather than guessing.

Live-data lookups (read-only). When the active provider is **Anthropic, OpenAI, or Google Gemini**, the assistant can additionally *look up* your live data to answer questions like *"What's Aceco's AR balance?"* or *"How many of item X do we have on hand?"*. These lookups are strictly **read-only** — the assistant itself cannot create, change, or delete anything — and they run under the signed-in user's account, so every question is rate-limited and recorded in the Audit Log. The other four providers (Groq, Cerebras, NVIDIA, OpenRouter) answer how-to questions only, with no data access — this restriction is deliberate, because weaker models are less reliable at using lookups correctly. See §8.14.2 for what can be asked.

Drafting transactions (you confirm). With the same three providers, the assistant can also *prepare* three kinds of transactions from a plain-English request — a **sales order**, a **customer (AR) payment**, and an **inventory adjustment**. The assistant only fills in a draft; you see a confirmation card in the chat and **nothing exists until you press Confirm**. The Confirm click submits the transaction under *your* user account, through the same permission checks, validation, credit checks, audit logging, and GL posting as the regular screens — if you don't have permission to create sales orders, Confirm is rejected exactly as the Sales Orders screen would reject you. See §8.14.2.

1.9.14.1 8.14.1 Setting It Up (Administrator)

The assistant can be powered by any of **seven providers** — Anthropic (Claude), OpenAI (GPT), Google Gemini, Groq, Cerebras, NVIDIA, or OpenRouter — configuration lives on its own page, **Admin → Setup → AI Setup**, not under Company Settings. You can store more than one model config (e.g., one Anthropic key and one Groq key) but only **one is active** at a time; that's the one the ▯ AI Help button uses. Note that **live-data lookups (§8.14.2) work only with Anthropic, OpenAI, and Gemini** — the other four providers answer how-to questions only.

1. Get an API key from the provider of your choice:

Provider	Where to get a key	Key looks like	Notes
Anthropic	console.anthropic.com	`sk-ant-…`	Paid (billing on file); best answer quality
OpenAI	platform.openai.com	`sk-…`	Paid (billing on file)
Google Gemini	aistudio.google.com	`AIza…`	Free tier available
Groq	console.groq.com	`gsk_…`	Very fast; free tier too small for this assistant — needs Dev Tier (see note below)
Cerebras	cloud.cerebras.ai	`csk-…`	Very fast; free tier has per-minute token limits (see note below)
NVIDIA	build.nvidia.com	`nvapi-…`	Free API credits for developers

Provider	Where to get a key	Key looks like	Notes
OpenRouter	openrouter.ai	`sk-or-v1-…`	One key, many models (incl. Claude/GPT); pay-as-you-go

> **Free-tier token limits.** Every question sends the full reference material (this guide plus the schema digest — roughly **50,000 tokens**) to the provider. Free tiers with small *tokens-per-minute* quotas cannot accept a request that large: Groq's free tier (6k–12k TPM depending on model) rejects every question with *"AI request failed: Request too large…"* — verified in testing — and requires their paid **Dev Tier**. Gemini's free tier (250k TPM on Flash models) is large enough. If you see a "Request too large" error, the key and setup are fine; the provider's tier is the limit.

2. In OpenERP go to **Admin → Setup → AI Setup** and click **+ Add Model**.
3. Pick the **Provider**, give it a **Label** (e.g. "Production Claude"), then pick a **Model** from the dropdown — it lists the current models for whichever provider you chose and always starts on the recommended one, so there's nothing to type or mistype. Pick **Other (type manually)…** only if you need a specific model not in the list (e.g. a brand-new release). Paste the **API Key** and **Save**.
4. In the grid, click the **Active** column (○) on the row you want live — it turns into a green ▯ and any previously active row is turned off automatically. The ▯ AI Help button appears for all users as soon as a row is active.

Click the ▯ on the active row again to turn AI Help off entirely (no active row = disabled) without deleting the configuration. API keys are stored server-side only and are never sent back to the browser in full — the grid and edit form always show a masked value (•••• plus the last 4 characters); leave that as-is when editing to keep the existing key, or paste a new one to replace it. Select any row to reveal **Edit** (change the label, model, or key) and **Delete** buttons below the grid.

Model choice. The dropdown's top option per provider is the recommended default (e.g. *Claude Opus 4.8* for Anthropic, *GPT-5.5* for OpenAI, *Gemini 2.5 Flash* for Google), with faster/cheaper tiers below it. Use **Other (type manually)…** for any model the provider serves that isn't listed — Groq, Cerebras, NVIDIA, and OpenRouter in particular add and retire hosted models frequently, so check the provider's model list if a preset returns a "model not found" error. Typical cost ranges from free (Gemini's free tier — the only free tier confirmed to fit the assistant's large per-question context) to a few cents per question — repeat questions within a few minutes are cheaper on Anthropic because the reference material is cache-eligible there.

Limits. Each user can ask up to 60 questions per hour. Every question is recorded in the Audit Log (Admin → Audit Log, table `ai_chat`) with the user and a snippet of the question, so usage is visible.

1.9.14.2 8.14.2 Using It

- Click ▯ **AI Help**, type a question, press **Enter** (or click **Send**).
- Follow-up questions keep the conversation context — you can ask *"and how do I void it?"* after asking about invoices.
- The assistant knows which page you're on and can tailor the answer to it.
- ⟲ clears the conversation; ▯ closes the panel (the conversation is kept until you log out or reload).

- If the button doesn't appear, the assistant hasn't been enabled — see 8.14.1.
- When you open the chat, the welcome screen lists example questions grouped into □ **How-to questions** and (when the active provider supports it) □ **Your live data** and □ **Draft transactions** — click an example to send it, or use it as a template (examples with a "…" placeholder fill the input box for you to complete). You can also just ask *"what can you do?"*.

Asking about your live data. With Anthropic, OpenAI, or Gemini active (see 8.14.1), the assistant can run these read-only lookups for you:

You can ask about	Examples
Customers	*"Find customers in Houston"*, *"What's Aceco's AR balance and credit limit?"*, *"Is customer X on credit hold?"*
Vendors	*"Search vendors named Wenzhou"*, *"How much do we owe vendor Y? Which bills are open?"*
Items & stock	*"Find items matching 'ball valve'"*, *"How many of item 10-370CTBALL21 are on hand?"* (on-hand, committed, on-order, ATP, by warehouse)
Orders	*"Show open purchase orders"*, *"List recent sales orders for customer Z"*

Things to know:

- **Read-only, always.** The assistant cannot create a transaction, change a record, or post anything — if you ask it to, it will point you to the right screen instead. (A future phase may add transaction creation with an explicit confirm step; today it is lookup-only.)
- **Name it precisely for best results.** If several records match (e.g. "valve" matches many items), the assistant lists the candidates and asks which one you meant — answering with the exact item or customer number gets you there fastest.
- **Answers show their work.** Below each answer that used live data you'll see a small *"□ Checked: …"* line listing which lookups ran. **If that line is absent, the answer did not come from your data** — treat any specific numbers in it with suspicion and ask again more precisely. Weaker/faster models occasionally "improvise" a plausible-looking answer instead of looking it up; this is why data access is limited to the three strongest providers.
- Lookup results are capped (top 10–20 rows) — ask a narrower question rather than "list all customers".
- Each answer may run several lookups, and every lookup re-sends the reference material, so data questions consume more provider tokens than how-to questions — relevant on free tiers (see the token-limits note in 8.14.1). On Anthropic this is much cheaper than it sounds: the reference material is cached after the first round, so extra lookup rounds only pay the (small) new tokens.

Drafting transactions — the Confirm card. Ask in plain English and the assistant prepares the transaction for you:

- *"Create a sales order for Aceco: 10 of VLV-100 at $32 each"*
- *"Record a $500 check payment from Aceco, check #10456"*
- *"Adjust ITEM-X by −2, damaged in handling"*

What happens next, every time:

1. The assistant looks up the customer/item to make sure it's real (asking you to pick when several match), fills in defaults (list price, standard cost, today's date, the open GL period, an auto-generated document number), and shows a □ **confirmation card** in the chat with every figure spelled out.
2. **Nothing has been created at this point.** The card has **Confirm** and **Cancel** buttons; the assistant's text will always say the draft is awaiting your review.
3. Press **Confirm** and the transaction is submitted *as you* — same permissions, validation, duplicate checks, credit-limit block, audit trail, and GL auto-posting as the regular screen. The card turns into a green □ with the new document number, or shows the exact error (e.g. over credit limit) if the system rejects it.
4. Press **Cancel** and the draft is discarded — nothing is created, nothing to clean up.

Three transaction types are supported today: **sales orders**, **AR customer payments** (created unapplied — apply to invoices under AR → Payments), and **inventory adjustments**. For anything else (editing, voiding, posting, applying payments, purchase orders…), the assistant will point you to the right screen instead. Warnings appear on the card *before* you confirm — e.g. if the order would push the customer over their credit limit, the card says so (and Confirm would be rejected).

1.9.14.3 8.14.3 Keeping Its Knowledge Current

The assistant's knowledge lives in two places on the server:

Source	What it covers	How to update
`USERGUIDE.md`	Everything in this guide	Edit the guide — picked up automatically on the next question
`docs/ai-context/*.md`	Schema digest + any extra notes you add	See below

The **schema digest** (`docs/ai-context/schema-digest.md`) is generated from the SQL schema files by a script. After adding or changing tables, regenerate it:

```
C:\php\php.exe F:\OpenERP\tools\generate_ai_schema_digest.php
```

It extracts table names, column names/types, and the comments already written in the DDL — structure only, no data — so there is nothing confidential in it. You can also drop additional Markdown files into `docs/ai-context/` (e.g. company-specific workflow notes, naming conventions, "who to call" lists) and the assistant will use them immediately; each file is included on every question.

1.9.15 8.15 Message Log — the Outbound Message Queue

Admin → Tools → Message Log shows every customer text (SMS / WhatsApp) and notification email the system has queued: who it was addressed to, the exact message text, when it was scheduled/sent/delivered, how many send attempts were made, and the precise provider error when something failed.

How the queue works. Features that message customers (order-status updates today; more later) don't send directly — they add a row to the queue. The **queue worker** then:

1. picks up due PENDING rows and sends them via the configured channel;
2. on a transient failure, retries with a 1 / 5 / 15-minute backoff, up to **Max Send Attempts** (Twilio settings, default 3), then marks the message FAILED with the reason;

3. after Twilio accepts a text, follows up to record **actual delivery** — statuses progress `PENDING → SENT → DELIVERED`, or end in `FAILED` with the carrier's error code (e.g. *30034: A2P campaign not yet approved*).

Running the worker. Three interchangeable ways — all run the same code:

How	When to use
▶ **Process Queue Now** button on the Message Log page	On demand; also the easiest way to test
The **OpenERP Worker Service** (`service\OpenERPSvc.exe` — see `service\README.md`), which runs `tools\run_message_queue.php` every minute	**Recommended** for unattended production use
`tools\run_message_queue.php` via Windows Task Scheduler / cron (every minute)	If you prefer not to install the service

Row actions. **View** opens the full detail (message text, provider SID, error). **Resend** re-queues a FAILED / CANCELLED / SKIPPED message with a fresh attempt counter (for SKIPPED rows that had no recipient, you'll be asked for one). **Cancel** stops a PENDING message before the worker picks it up. Messages that were sent are never deleted — the log is the audit trail of what each customer was told.

1.9.16 8.16 Database Backups

Admin → Setup → Company → Database Backups — automatic nightly zip archives of your entire database plus all uploaded documents, with retention pruning and a run log. This is the difference between a bad day and a catastrophe: hardware fails, files get corrupted, people make mistakes. Configure it on day one.

How it works. On schedule, OpenERP asks the Advantage server for a **hot snapshot** of the data dictionary and every table (`sp_BackupDatabase`) — this is safe while users are connected and working; the snapshot is consistent as of the moment it started. The snapshot lands in the archive under `database\`, your uploaded documents (vendor/customer/part files) under `uploads\`, and the whole thing is written as one zip file, integrity-checked, then old archives beyond your keep-count are deleted.

Configuration (all in the Database Backups card; click **Save Changes** after editing):

Setting	Meaning
Backups Enabled	Master switch for scheduled runs (Run Backup Now works regardless)
Destination Folder	Where the zips go. A real path on the server — ideally a **different physical disk** than the data, or a network share. Never a mapped/subst drive letter
Schedule / Days	Daily with any subset of weekdays, or Weekly on one day
Start Time	The backup fires on the first check at or after this time — pick a quiet hour (2:00 AM default)
Backups to Keep	Retention: oldest archives beyond this count are deleted after each successful backup

Setting	Meaning
Source Folder	Leave blank — the default (the data folder) is right for almost everyone

What actually runs it. The schedule is executed by the **OpenERP Worker Service** (§8.11 / `service\README.md`), which checks every few minutes whether a backup is due. No service installed = no scheduled backups (the **Run Backup Now** button still works — it runs the backup immediately through the web server). Every attempt — scheduled or manual — appears in the **Recent backups** table with its status, size, file count, and archive name; failures show the error text.

Restoring. A backup is only as good as its restore. To restore:

1. Unzip the archive somewhere on the server (e.g. `C:\restore\`).
2. In any SQL tool connected to Advantage (or a one-line PHP script), run: `EXECUTE PROCEDURE sp_RestoreDatabase('C:\restore\database\openerp.add', '', 'E:\OpenERP\data\openerp.add', NULL);` — the destination must be a **new/empty** location; never restore on top of a live database you might still need. Indexes are rebuilt automatically during restore.
3. Copy the archive's `uploads\` folder back into the restored data folder.
4. Point `backend\php\config\database.local.php` (dd_path) at the restored `.add` if you're switching over to it.

Do a test restore once after setting backups up, and again after any major upgrade — five minutes of practice while calm beats learning the procedure during an outage.

The FILE fallback. If the hot snapshot isn't possible (the connecting user lacks backup rights, or the ADS server runs on a different machine than the web server and can't see the staging folder), the runner falls back to copying the raw data files into the zip. The run log shows Method = FILE, and files the database server had open at that moment are skipped (the note column says how many). A FILE backup of an in-use database is best-effort — if you see FILE instead of HOT, fix the cause (re-run `tools\create_dd.php`, which grants the backup right) rather than living with it.

1.10 9. Integration

This section covers three ways data can move between OpenERP and the outside world without anyone re-typing it: **EDI** (structured business documents exchanged with trading partners), **webhooks** (another system pushing data to OpenERP the instant something happens), and **recurring POs** (OpenERP automatically re-issuing the same purchase order on a schedule). The first two are about connecting OpenERP to *other companies' computer systems*; the third is about OpenERP automating a routine task for you internally.

1.10.1 9.1 EDI — What It Is and How Inbound Orders Arrive

EDI stands for Electronic Data Interchange. It's a decades-old but still very widely used standard for two companies' computer systems to exchange business documents — purchase orders, shipping notices, invoices — as structured text files, instead of a person emailing a PDF or faxing a paper order that someone else re-types. Large retailers, distributors, and manufacturers commonly *require* their suppliers to support EDI before they'll do business with them at all.

Each type of document has a standard three-digit code (this standard is called **X12**, the American format most common in the US):

Code	Name	Plain-language meaning
850	Purchase Order	"Here's an order — please fulfill it." Sent by the buyer to the seller.
856	Advance Ship Notice (ASN)	"Here's what we just shipped you, and when it'll arrive." Sent by the seller to the buyer after shipping.
810	Invoice	"Here's the bill for what we shipped." Sent by the seller to the buyer after invoicing.

Important limitation to know about today's implementation: when OpenERP receives an inbound 850, it currently creates a **Purchase Order** against **the first active vendor** in your vendor list — it does not yet look at *which* trading partner sent the document to pick the right vendor. This works fine if you only receive EDI from one source, but if you're setting this up for multiple trading partners, verify the vendor on each resulting PO before acting on it until this is refined.

1.10.1.1 How an inbound EDI document actually reaches OpenERP

This is the part that trips people up: **OpenERP does not "listen" on the internet for EDI the way it does for its own web pages.** A real trading partner's EDI system doesn't call OpenERP directly. Real-world EDI almost always travels through one of these:

- **A VAN (Value-Added Network)** — a third-party mailbox/clearinghouse service (e.g. SPS Commerce, TrueCommerce, Edicom) that both companies connect to. Your partner drops the document in your VAN mailbox; you retrieve it from there.
- **AS2** — a secure, direct point-to-point internet connection standard, common with large retailers (Walmart, Home Depot, etc.) who require it.
- **SFTP** — a simple secure file-drop folder, less common for retail but still used.

OpenERP's role is only the **last step**: it exposes one API call, `POST /edi/inbound/850`, that accepts an already-received X12 document as text and turns it into a Purchase Order. Getting the document *from* the trading partner *to* that API call is a separate piece you (or your EDI provider) have to set up — typically a small script or a feature of your VAN/AS2 software that watches for new documents and forwards each one to OpenERP with an HTTP call, or an in-house scheduled script that logs into the VAN's SFTP/mailbox, downloads new files, and posts them.

1.10.1.2 What information is needed to set it up

1. **Register the trading partner** — go to **Admin → EDI Transactions → Manage Partners** and add: a short internal Code, the partner's Name, the ISA Qualifier and ISA Sender ID they use (these identify who sent a document — get these from your partner's EDI setup instructions or your VAN), and Direction (inbound only / outbound only / both).
2. **Get your VAN, AS2 tool, or forwarding script configured to call**:

   ```
   POST /edi/inbound/850
   Authorization: Bearer <a valid OpenERP login token — see below>
   ```

```
Content-Type: application/json
Body: { "partner_id": "<the edi_partner_id from step 1, optional>", "x12": "<the full raw X12 d
```

Note the field is named **x12**, containing the *entire* raw X12 document as one string (segment terminators and all) — not a parsed/structured version of it.

3. **Authentication:** unlike a public webhook, this endpoint requires a logged-in OpenERP session, because inbound EDI is expected to come from a trusted internal integration (your VAN-forwarding script), not directly from a random external server. Whatever script or tool calls this endpoint needs to first call `POST /auth/login` with a service account's username/password to get a token, then include `Authorization: Bearer <token>` on the `/edi/inbound/850` call.

1.10.1.3 How to test it manually

You can hand-build a tiny X12 850 and post it directly to confirm your setup works before wiring up a real VAN connection. The parser figures out the field/segment separators from the fixed positions in the very first segment (`ISA`), so a minimal valid document looks like:

```
ISA*00*          *00*          *ZZ*TESTPARTNER    *ZZ*OPENERP        *230101*1200*U*00501*000000001*0*T*>~
TEST-001**20230101~PO1*1*3*EA*10.50**BP*ITEM-NUMBER-HERE~SE*4*0001~GE*1*1~IEA*1*000000001~
```

The `BEG` segment's third field (`PO-TEST-001` above) becomes your PO's number; each `PO1` segment becomes one PO line, matched to your item master by item number (7th field). Results — success or failure — always appear in **Admin → EDI Transactions**, whether triggered by a real partner or a manual test.

1.10.2 9.2 EDI — Outbound: Shipping Notices (856) and Invoices (810)

Outbound works the other direction, and is simpler because *you* control when it happens:

```
GET /edi/outbound/856/{so_id}     → generates an X12 856 (ASN) for a shipped Sales Order
GET /edi/outbound/810/{invoice_id} → generates an X12 810 (Invoice) for an AR Invoice
```

Visiting either URL (with a valid login) downloads a ready-to-send X12 text file built from that specific order or invoice's real data.

Routing it to the trading partner is, again, not something OpenERP does for you. Just like inbound, OpenERP's job stops at *generating a correctly-formatted document* — actually transmitting it requires the same kind of VAN, AS2, or SFTP connection used for inbound, just in reverse: you (or your EDI provider's software) take the generated file and drop it into your VAN's outbound mailbox, or push it through your AS2 software, addressed using the ISA/GS identifiers you set up for that partner. In practice this is usually automated too: after a shipment or invoice is created, a script (yours or your EDI provider's) calls the download URL and hands the result to your VAN/AS2 tool.

1.10.3 9.3 Webhooks — Letting Another System Push Data Into OpenERP

A webhook is the opposite of the usual "go check for updates" pattern. Instead of OpenERP repeatedly asking your e-commerce platform "any new orders yet?", you configure your e-commerce platform to immediately call a specific OpenERP web address the moment a new order happens — it pushes the order to you in real time, the instant it's placed, instead of you polling for it.

1.10.3.1 How to reach OpenERP with a webhook

Configure your e-commerce platform (Shopify, WooCommerce, or any custom storefront) to send an HTTP POST to:

```
POST https://<your-openerp-server>/ecomm/webhook
Content-Type: application/json
```

No login/authentication token is required on this one endpoint — that's what makes it a true webhook, since your e-commerce platform has no OpenERP account of its own to log in with. Because of that, treat the webhook URL itself as something to keep semi-private, and if your platform supports it, restrict which IP addresses are allowed to reach this endpoint at your firewall/reverse proxy.

1.10.3.2 What payload OpenERP expects

```json
{
  "source": "SHOPIFY",
  "external_id": "order_12345",
  "customer": {
    "email": "buyer@example.com",
    "name": "Jane Buyer",
    "address1": "123 Main St",
    "city": "Springfield",
    "state": "IL",
    "zip": "62704"
  },
  "lines": [
    { "item_number": "WIDGET-A", "qty": 2, "price": 9.99 }
  ]
}
```

- `source` — a short label for which platform sent it (used for filtering in the order list, and for duplicate detection — see below).
- `external_id` — that platform's own order number/ID. Required.
- `customer.email` — required; OpenERP looks up an existing customer by this email, or creates a new one automatically if none matches.
- `lines[].item_number` — matched against your item master; if no match is found, the line is still created using whatever `description` (optional field) you send instead.

1.10.3.3 What happens when it fires

OpenERP finds-or-creates the customer, creates a new Sales Order with the given lines, and records the import under **Admin → E-Commerce Import** with a status of Imported, Pending, or Error (with the reason, if one failed). **It's safe for your e-commerce platform to send the same webhook call twice** (a common real-world occurrence — platforms often retry on any hint of a failed delivery): OpenERP checks `source` + `external_id` together, and if that combination was already successfully imported, it returns a "duplicate" response and does not create a second Sales Order.

1.10.4 9.4 Recurring Purchase Orders

1.10.4.1 Why a business would want this

Think of anything you order from the same vendor, in roughly the same quantity, on a predictable schedule — shop supplies, cleaning contract materials, a raw material a manufacturer buys every week, office consumables reordered monthly. Today, without this feature, someone in Purchasing has to remember to do it, recall (or look up) the vendor, remember every line item and quantity, re-type it all into a new PO, and hope they didn't fat-finger a quantity or forget an item. Miss it entirely one cycle, and a production line or a shop can run out of something basic.

A Recurring PO Template solves this by letting you build the order once and re-issue it either with one click or fully automatically, on whatever schedule you choose — weekly, bi-weekly, monthly, quarterly, or annually. Each time it runs, it advances itself to the next due date, so it's always ready for the following cycle without anyone touching it.

1.10.4.2 How to set it up

Go to **Purchasing → Recurring POs → + New Template** and fill in: - **Template Code** — a short internal name (e.g. `SHOP-SUPPLIES-MONTHLY`) - **Vendor**, **Terms**, **Warehouse** (optional) — same as a normal PO - **Frequency** — how often it should re-fire - **Next Run Date** — the first date it becomes due - **Lines** — the items, quantities, and unit costs that make up the standing order

1.10.4.3 Running it — by hand or automatically

- **By hand:** click **Run Now** on a due template. This immediately creates a real, normal Purchase Order — it shows up in your regular PO list, gets received and billed exactly like any manually-created PO — and advances the template's Next Run Date forward by its frequency.
- **Fully automatically:** run this command (it finds every active template whose Next Run Date has arrived, generates a PO from each, and advances each one):

  ```
  C:\php\php.exe -c C:\php\php_sapads.ini C:\OpenERP\tools\run_po_recurring.php
  ```

  Just like the Scheduled Reports runner in §8.11, this command does nothing on its own until something triggers it on a schedule. The **OpenERP Worker Service** (see §8.11) already ships configured to run it daily at 6:00 AM — install the service once and this is covered. Alternatively, set it up in **Windows Task Scheduler**:

  1. **Create Task...** → name it "OpenERP Recurring POs."
  2. **Triggers → New:** Daily (or however often you want it checked — running it more often than any of your templates' frequencies is harmless, since it only acts on templates that are actually due).
  3. **Actions → New:** Program `C:\php\php.exe`, arguments `-c C:\php\php_sapads.ini C:\OpenERP\tools\run_po_recurring.php`.

  From then on, a recurring order for shop supplies (or anything else you template) generates itself right on schedule — nobody has to remember it.

1.11 10. Management Operating Procedures

This section describes the recommended business routines for running the system efficiently. These are operating procedures, not software instructions — they describe *when* and *why* to perform tasks, not just *how*.

1.11.1 10.0 The Analytics Section (Pillar Dashboards)

The **Analytics** sidebar section holds one executive dashboard per master-record pillar. Each combines KPI tiles, charts, and a detail table. Check them weekly (or before customer/vendor reviews) rather than daily:

- **Inquiry Dashboard** — the business-development pipeline. Tiles: open pipeline count and potential value (line qty × target price for DRAFT/QUOTING inquiries), **win rate** (awarded ÷ (awarded + lost) — cancelled inquiries don't count against you), awarded value, and average days from receipt to award. Charts: inquiries by status, monthly received/awarded/lost. Table: most active customers with per-customer win rate. A falling win rate or an aging pipeline is the early signal to review pricing or supplier response times.
- **Customer Dashboard** — relationship health. Tiles: revenue for the selected invoice-date range, **AR exposure** (open + partial invoice balances), open SO backlog, and customer quote win rate (with an on-credit-hold badge when any customer is held). Chart: top customers by revenue. Table: per-customer revenue, AR balance (amber when nonzero), and open SO value — customer concentration is visible at a glance.
- **Part Dashboard** — the production portfolio. Tiles: active parts, **engineered parts** (carrying a drawing number), preferred-vendor coverage, and **single-source risk** (engineered parts with only one vendor in the catalog — the tile turns amber when nonzero; find a backup supplier). Charts: parts by manufacturing method, top parts by shipped revenue. Table: top parts with drawing numbers.
- **Vendor Scorecard** — supplier performance (§4.1); linked here so all four pillars' dashboards live in one place.

  Charts blank after an upgrade? The dashboards use the Chart.js library, which is installed automatically by `setup.ps1` (or `npm install` in the `frontend` folder). If the KPI tiles show numbers but the chart areas are empty — with an import error in the browser console — the frontend dependencies were not refreshed after the upgrade: re-run `setup.ps1` (or `npm install`) and restart the frontend. See INSTALL.md Step 9.

1.11.2 10.1 Daily Routines

1.11.2.1 Finance / Controller

- **Check the Dashboard.** Review cash, open AR, open AP, and overdue AR tiles. Any sharp movements overnight indicate transactions that need review.
- **Process AR receipts.** Enter payments received by mail or wire from the prior day into **AR → Payments** and apply them to open invoices.
- **Process AP bills.** Enter vendor invoices received by mail or email into **AP → Bills**. Code each bill to the correct period and GL account.
- **Review the Audit Log** (if anomalies are suspected). Filter by yesterday's date to see all transactions.

1.11.2.2 Warehouse / Receiving

- **Receive open POs.** For each delivery that arrives, locate the PO in **Purchasing → Orders** and click **Receive**. Enter the actual quantities received. Print the receipt for the packing file.
- **Check reorder suggestions.** After receiving, review **Inventory → Reorder Suggestions** to see if any items have dropped below reorder point. Notify purchasing.

1.11.2.3 Sales / Customer Service

- **Check the back-order report.** Review **Sales → Back-Orders** for any lines with a past ship date. Contact customers about delays and update promise dates.
- **Process new orders.** Enter customer orders promptly — the ATP check gives the customer real-time availability.

1.11.3 10.2 Weekly Routines

1.11.3.1 AR Collections (every Monday, or first business day of the week)

1. Run **AR → Aging**. Sort by the 61–90 and 91+ columns descending.
2. For every customer with a balance in the 61–90 column: send a statement (AR → Statements → Email) or make a phone call. Log the contact in Collections.
3. For every customer with a balance in the 91+ column: escalate — contact the customer directly, involve the sales rep, and consider placing the account on credit hold (AR → Customers → Hold).
4. For customers on hold, the system blocks new orders automatically. Release the hold (AR → Customers → Release) only when payment is received or a payment plan is agreed.
5. Discuss write-off candidates with management. Write-offs over a threshold (e.g., $1,000) should require controller approval.

1.11.3.2 AP Check Run (typically Wednesday or Thursday)

1. Run **AP → Aging**. Identify all bills due within the next 7 days.
2. Add bills that offer an early-pay discount (e.g., 2/10-NET30) if the discount is worth more than your cost of funds.
3. Create payments in **AP → Payments** for each vendor.
4. Print checks or initiate ACH/wire transfers through your bank.
5. After the bank processes the payments, match them in **Bank Reconciliation** when the statement updates.

1.11.3.3 Purchasing Review

1. Review open POs past their expected delivery date (**Purchasing → Orders**, filter OPEN/PARTIAL, sort by need-by date).
2. Contact vendors on overdue deliveries and update the expected date.
3. Review **Inventory → Reorder Suggestions** for any new shortfalls.
4. Process any RFQs that have received all vendor responses — compare and convert the winner to a PO.

1.11.3.4 Sales Commission Review

1. Run **Sales → Commissions** for the current month-to-date.
2. Share with sales management to track rep performance against quota.
3. Commissions accrue on invoiced SOs only (not on orders or shipments) — remind reps to push customers to accept invoices promptly.

1.11.4 10.3 Monthly Period-End Checklist

Perform these steps in order, on or before the last business day of each month. Do not close the period until all steps are complete.

Step 1 — Complete all transactions for the period. - Ensure all AR invoices for the month are entered and emailed. - Ensure all AP bills received during the month are entered. - Ensure all PO receipts for the month have been received in the system. - Ensure all cash receipts received during the month are applied. - Ensure all AP payments issued during the month are recorded.

Step 2 — Inventory. - Post any pending cycle count approvals. - Post any pending physical count sessions. - Run landed cost allocation for any containers that arrived during the month (**Purchasing → Landed Costs**). - Review **Inventory → Valuation (FIFO)** — does the total inventory value look reasonable?

Step 3 — GL Adjustments. - Run all recurring journal entries for the month (**GL → Recurring JEs**). - Run depreciation for all fixed assets (**GL → Fixed Assets → Depreciate** for each asset). - Post any accruals required by your accountant (accrued wages, accrued expenses, deferred revenue). - If you have foreign currency balances, run **GL → FX Revaluation**.

Step 4 — Reconcile. - Reconcile the bank account in **Financials → Bank Reconciliation** for the month just ended. - Confirm the GL Cash balance matches: Bank Statement Balance + Outstanding Deposits − Outstanding Checks. - Reconcile the AR subledger to the GL: the sum of open AR invoice balances should equal the GL AR account balance. - Reconcile the AP subledger to the GL: the sum of open AP bill balances should equal the GL AP account balance.

Step 5 — Review financial statements. - Run the **Trial Balance** for the period. Confirm it balances. Investigate any unusual account balances. - Run the **Income Statement**. Compare to prior month and prior year same month. Explain significant variances. - Run the **Balance Sheet**. Confirm AR, AP, and Inventory totals match the reconciliations in Step 4.

Step 6 — Close the period. - Once satisfied that the period is correct, go to **GL → Periods** and click **Close** for the month. - A closed period cannot receive new transactions. If a correction is needed after closing, it must be posted in the next open period with a reversing entry pair.

Step 7 — Distribute reports. - Send the Income Statement and Balance Sheet to management. - Send the AR Aging to the collections team. - Send the AP Aging to the controller / CFO. - Send the Sales Commission report to the sales manager. - Send the Inventory Valuation summary to the operations manager.

1.11.5 10.4 Quarterly Procedures

Sales Tax Filing: - Run **Admin → Sales Tax** reports for the quarter. - Calculate tax collected by jurisdiction and remit to the applicable state/local tax authorities. - File sales tax returns as required by each jurisdiction.

Inventory ABC Review: - Run **Inventory → ABC Classify** to re-rank items by actual COGS for the quarter. - Adjust cycle count frequencies based on the new ABC classification. - Review reorder points for A items — these should be rechecked against current demand patterns.

Vendor Performance Review: - Pull **Purchasing → 3-Way Match** with the Exception filter to find vendors with chronic over-billing or over-receiving issues. - Review **Purchasing → Inbound Shipments** for vendors with chronic late deliveries. - Use this data in vendor negotiations.

Budget vs Actual: - Run **GL → Budgets → Budget vs Actual** for the quarter. - Identify accounts that are significantly over or under budget. - Adjust forecasts for the remainder of the year and communicate to department heads.

1.11.6 10.5 Annual Procedures

1.11.6.1 Year-End Financial Close

1. **Complete all December transactions** using the Monthly Period-End Checklist (§10.3).
2. **Physical inventory count** — schedule and complete a full physical count in the last week of December. Post all variances.
3. **Depreciation** — run December depreciation for all fixed assets.
4. **Accruals** — consult your accountant for year-end accruals:
   - Accrued bonus/vacation
   - Prepaid expenses to amortize
   - Deferred revenue
5. **Close all 12 periods** in the fiscal year.
6. **Run Year-End Close** — go to **GL → Year-End Close**, review the summary, and click Run Close. This zeros income statement accounts and credits net income to Retained Earnings.
7. **Create the new fiscal year** and all 12 periods.
8. **Verify opening balances** — run the Balance Sheet for the new year's first period. All balance sheet accounts should show the same closing balance as December 31.

1.11.6.2 1099 Preparation (US)

1. In January, run **AP → 1099 Report** for the prior calendar year with a minimum of $600.
2. For each vendor with a ▯ badge (missing Tax ID), contact the vendor and obtain a completed W-9.
3. Enter the Tax ID on the vendor record in **AP → Vendors**.
4. Print or e-file 1099-NEC forms. Due to recipients: **January 31**. Due to IRS: **January 31** (electronic) or **February 28** (paper).

1.11.6.3 Price List and Contract Review

1. Run a margin analysis by item category — pull Sales data vs. FIFO cost from the Inventory Valuation report.
2. Identify items where your cost has risen but list prices have not been updated.
3. Update price lists and quantity breaks in **Sales → Price Lists**.
4. Review and renew (or renegotiate) customer contracts in **Sales → Contracts**.

1.11.6.4 Sales Rep Commission Reconciliation

- Pull the full-year **Sales → Commissions** report.
- Reconcile to any commission statements you have paid manually during the year.
- This is the basis for annual commission W-2 or 1099 reporting.

1.11.7 10.6 Exception Handling

1.11.7.1 Customer Disputes a Bill

1. Review the original SO, ship confirmation, and AR invoice — confirm quantities and pricing.
2. If the customer is correct: issue a **Credit Memo** (AR → Credit Memos) for the disputed amount and apply it to the invoice.
3. If the customer is wrong: send a copy of the original documentation and the invoice PDF. Log the dispute in the Collections notes.
4. Never adjust an invoice after it has been sent — create a credit memo instead. Changing a posted invoice creates audit issues.

1.11.7.2 Vendor Overcharges on a Bill

1. Do not pay the overcharged amount.
2. Contact the vendor and request a corrected invoice or a credit memo.
3. If the vendor issues a credit memo, enter it as an **AP Debit Memo** (**AP → Debit Memos**) and apply it against the bill.
4. If the vendor refuses and the amount is material, void the original bill, enter the correct amount as a new bill, and document the situation in the notes.

1.11.7.3 Inventory Discrepancy

1. Identify the item, warehouse, and bin.
2. Count the actual quantity on hand.
3. Review recent inventory transactions in the Audit Log — look for unauthorized adjustments or receiving errors.
4. If the count is correct and the system is wrong: post an adjustment in **Inventory → Items → Adjust** with a note explaining the investigation.
5. If the discrepancy is large or unexplained: investigate before adjusting — unexplained shrinkage may indicate theft.

1.11.7.4 Bank NSF (Returned Check)

1. The bank will debit your account for the returned check amount.
2. In the system, void the original AR payment (**AR → Payments → Void**). This reverses the application and restores the invoice balance.
3. Post a manual journal: DR Bank Charges / CR Cash for the NSF fee the bank charges you.
4. Contact the customer immediately. Place their account on credit hold pending resolution.
5. Re-invoice the customer for the NSF fee if your policy allows.

1.12 11. Reference

1.12.1 11.1 Document Status Codes

Sales Orders and Quotes: `DRAFT` → `SENT` → `ACCEPTED` / `EXPIRED` / `CANCELLED` (quotes) `OPEN` → `PARTIAL` → `SHIPPED` → `INVOICED` / `CANCELLED` (orders)

Purchase Orders and Blanket POs: `OPEN` → `PARTIAL` → `RECEIVED` → `BILLED` / `CANCELLED`

Purchase Requisitions: `DRAFT` → `PENDING` → `APPROVED` / `REJECTED` → `CONVERTED` / `CANCELLED`

RFQs: `DRAFT` → `SENT` → `CLOSED` / `CONVERTED`

RMAs and Vendor RMAs: `PENDING` → `APPROVED` → `RECEIVED` → `CREDITED` / `CLOSED` / `CANCELLED`

AP Bills / AR Invoices: `OPEN` → `PARTIAL` → `PAID` / `VOID`

GL Batches: `OPEN` → `POSTED`

Inbound Shipments (containers): `PENDING` → `IN_TRANSIT` → `ARRIVED` → `CLEARED` / `HELD`

1.12.2 11.2 PDF Generation

Every major document has a **PDF** button or link:

Document	Endpoint
AR Invoice	GET /ar/invoices/{id}/pdf
Customer Statement	GET /ar/customers/{id}/statement/pdf
Sales Order / Packing List	GET /so/{id}/pdf
Purchase Order	GET /po/{id}/pdf
Certificate of Conformance	GET /certs/{id}/pdf

PDFs embed the company logo, configurable header subtitle, footer note, and per-document terms. Customize these under **Admin → Company Settings → Document Templates**.

1.12.3 11.3 GL Auto-Posting Summary

The following events automatically post a GL batch:

Event	Debit	Credit
AR Invoice created	AR	Revenue (+ Tax Payable if tax > 0)
AR Payment applied	Cash	AR
AP Bill created	Expense	AP
AP Payment applied	AP	Cash
SO Shipment	COGS	Inventory
Inventory Adjustment (positive)	Inventory	Inv-Adj
Inventory Adjustment (negative)	Inv-Adj	Inventory
AR Credit Memo created	Revenue	AR
AP Debit Memo created	AP	Expense
AR Invoice void	Revenue	AR (reversal)
AR Payment void	AR	Cash (reversal)
AP Bill void	AP	Expense (reversal)
AP Payment void	Cash	AP (reversal)
Deposit applied to invoice	Deposits liability	AR
Year-end close	Revenue / Expense	Retained Earnings
Monthly depreciation	Depreciation Expense	Accum. Depreciation
FX gain on AP payment	AP	FX Gain
FX loss on AP payment	FX Loss	AP

Account resolution order: company_config key → first active non-header account with matching subtype → error logged, transaction still committed.

1.12.4 11.4 API Base URL

The REST API is at http://<server>:8081. All endpoints require Authorization: Bearer <token> except POST /auth/login.

Common patterns: - GET /ar/customers?page=1&per_page=50&search=acme — paginated list with search - POST /ar/customers — create - PUT /ar/customers/{id} — update - GET /ar/customers/{id} — single record

Paginated responses include total, page, per_page, and data fields.

1.12.5 11.5 Diagnostic Tools

Located in the tools/ directory (run with C:\php\php.exe -c C:\php\php_sapads.ini tools\<script>.php):

Script	Purpose
`install.php`	Interactive first-run installer / configuration wizard
`create_dd.php`	Create or refresh the SAP ACE data dictionary
`seed.php`	Seed payment terms, tax codes, and chart of accounts
`seed_auth.php`	Seed permissions, groups, and admin user
`seed_demo.php`	Load a full demo dataset (Meridian Industrial Supply)
`setup_users.php`	Create the `openerp_app` ADS user and lock down the DD
`smoke_test.php`	End-to-end HTTP smoke test (42 checks)
`rbac_test.php`	RBAC validation (32 checks)
`run_scheduled_reports.php`	Execute due scheduled reports and email results

1.12.6 11.6 Primary Key Format

All record IDs are 13-character strings in the format `MMYYnnnnnnnn`: - `MM` — 2-character table mnemonic (e.g. `AR` for invoices, `SO` for sales orders) - `YY` — 2-digit year - `nnnnnnnn` — 8-digit zero-padded sequence number

Example: `AR26-00000042` is the 42nd AR invoice record created in 2026. IDs are generated by the `NewSeqKey()` database function.

1.12.7 11.7 Session and Security Notes

- Tokens expire after **8 hours**. The frontend redirects to login automatically.
- **Logout** (`POST /auth/logout`) immediately invalidates all sessions for that user — other browser tabs will be signed out on their next API call.
- **Password change** and **admin force-logout** also invalidate sessions.
- All traffic should run over HTTPS in production. Configure a reverse proxy (nginx, IIS ARR) to terminate TLS in front of port 8081.
- The data dictionary is locked (`ADS_DD_LOG_IN_REQUIRED`) so anonymous connections are rejected at the database level. The backend connects as the `openerp_app` ADS user.

1.12.8 11.8 Internal Controls Checklist

These are basic controls a controller or auditor would expect to find. Review periodically:

Control	How to verify in OpenERP
No single employee can create and apply an AR payment	AR staff should have AR access; only the controller approves write-offs
All AP bills have a matching PO and receipt	Run 3-Way Match report monthly; investigate exceptions
Inventory adjustments are documented	Audit Log → filter by table `inventory_transactions` and action `ADJUST`
Terminated employees are immediately blocked	Admin → Users → Deactivate on last day of employment

Control	How to verify in OpenERP
Bank accounts are reconciled monthly	Financials → Bank Reconciliation; confirm cleared dates match statement
Period is closed after month-end review	GL → Periods; all prior periods should show CLOSED status
Physical count performed at least annually	Physical Count sessions; post within 5 business days of count
Financial statements reviewed by management monthly	GL → Reports; Trail Balance, Income Statement, Balance Sheet